

Remaining corrections — final

Leader lines that do not end on the structure they name

Rounds Codex · compiled 16 August 2026

HOW TO READ EVERY FIGURE IN THIS DOCUMENT

- RED** crossed circle — where the leader, dot or badge ends NOW. The number matches the table.
- GREEN** plain circle — where it belongs, with an arrow running from red to green.

Every coordinate is in the SHIPPED page's own pixels, origin top-left. Where a page is off-standard its size is printed in the page header, so the numbers can be scaled if you work from a larger master.

Where a target is given as a coordinate it was measured, and the tissue under it was sampled with the leader's own ink excluded. Where a target is described in words but not numbered, it could not be pinned to a pixel from the shipped render and needs placing from the source file — those are marked PLACE FROM SOURCE and are not guesses to be actioned blindly.

Some rows cannot be fixed by moving anything: the structure the label names is not drawn on the page. Those say so explicitly. They need artwork or a different view, and re-aiming the leader would only move the error.

WHAT IS IN HERE

25 gallery pages, 149 measured findings. These are the pages whose corrections are measured and still open. Pages already corrected and deployed are not included.

PAGE	FINDINGS	WHAT IT IS
hyponatremia — page 2	3	ADH production & release; nephron anatomy
bipolar — page 2	4	Neuroanatomy & neurobiology — circuit figure
b12-anemia — page 2	4	Labeled anatomy: vitamin B12 & folate absorp
cirrhosis — page 2	2	Normal liver anatomy & portal venous system
covid — page 2	3	Respiratory tract & viral entry
cholecystitis-p2	7	Anatomy Overview: Hepatobiliary System
croup-p2	4	Anatomy: Pediatric Airway
fracture-p2	7	Anatomy: Normal Bone & Common Fracture Locat
gi-bleed-p2	4	Upper GI Anatomy: Structures at Risk for Ble
influenza-p2	6	Labeled Respiratory Anatomy
leukemia-p2	6	Normal Bone Marrow & Hematopoiesis Anatomy
lung-cancer-p2	7	Labeled Anatomy & Typical Tumor Locations
lymphoma-p2	6	Anatomy: Lymphatic System
meningitis-p2	12	Anatomy of the Meninges & CSF Spaces

metabolic-syndrome-p1	7	Normal Anatomy & Visceral Adiposity
ms-p2	8	Labeled CNS Anatomy
myasthenia-p2	4	Labeled Neuromuscular Junction Anatomy
osa-p2	6	Upper Airway Anatomy – Lateral & Cross-Section
osteoarthritis-p2	7	Anatomy & Joint Structure
ra-p2	7	Anatomy: The Normal Joint
sickle-cell-p2	5	Anatomy: Hemoglobin, RBCs, Spleen & Microvasculature
t2dm-p2	6	Normal Pancreatic & Endocrine Anatomy
thrombocytopenia-p2	5	Anatomy & Physiology of Platelet Production
thyroidstorm-p2	10	Anatomy & Physiology
tia-p2	9	Cerebral & Carotid Vascular Anatomy

How to make the corrections

1. Moving a leader tip or a dot — most of this document

Erase the last segment along the LEADER'S OWN AXIS, then redraw to the green point. Two things keep it invisible: take the stroke width and colour by sampling the leader itself a little way back from the tip, and erase along the axis rather than with a box, so the surrounding artwork is never touched.

The test that the move is clean: at the OLD endpoint, ink in a 13x13 box should fall to ZERO. If ink remains there, the leader was extended rather than moved, and the page now carries two terminators.

2. Never repaint artwork

If a fix would mean painting over illustration - erasing a disc of shaded tissue, rebuilding a background, redrawing a structure - stop and mark the page for a re-render instead. This has been attempted here and it does not repair cleanly; every large tissue-side erase left a visible patch. Several rows in this document say exactly that and are addressed to the illustrator, not to Photoshop.

3. Rows that no move can fix

Where the structure the label names is NOT DRAWN on the page, re-aiming the leader only moves the error somewhere else. Those rows say so in the text and need artwork or a different view - for example a lateral inset, an added slab, or a redrawn tracing. Rows marked PLACE FROM SOURCE are ones we could not pin to a pixel from the shipped render; place them from the layered file.

4. Save at the SAME pixel dimensions

Do not resample, resize, crop or rotate. The coordinates in this document are in the shipped page's own pixels, and the app, the thumbnails and the download PDF are all built from that geometry. Save as JPEG at high quality - we re-encode on integration at the file's own quantisation tables and chroma sampling, so an exact quality match on your side is not needed, but a resize breaks everything downstream.

5. Send the corrected page back; do not rebuild anything else

We regenerate the thumbnail and rebuild that gallery's download PDF. That second step matters: the download PDF is a separate build artefact and drifts from the pages if it is not rebuilt - four galleries shipped for months with an old logo in the PDF while the app showed the current one.

6. One page at a time is fine

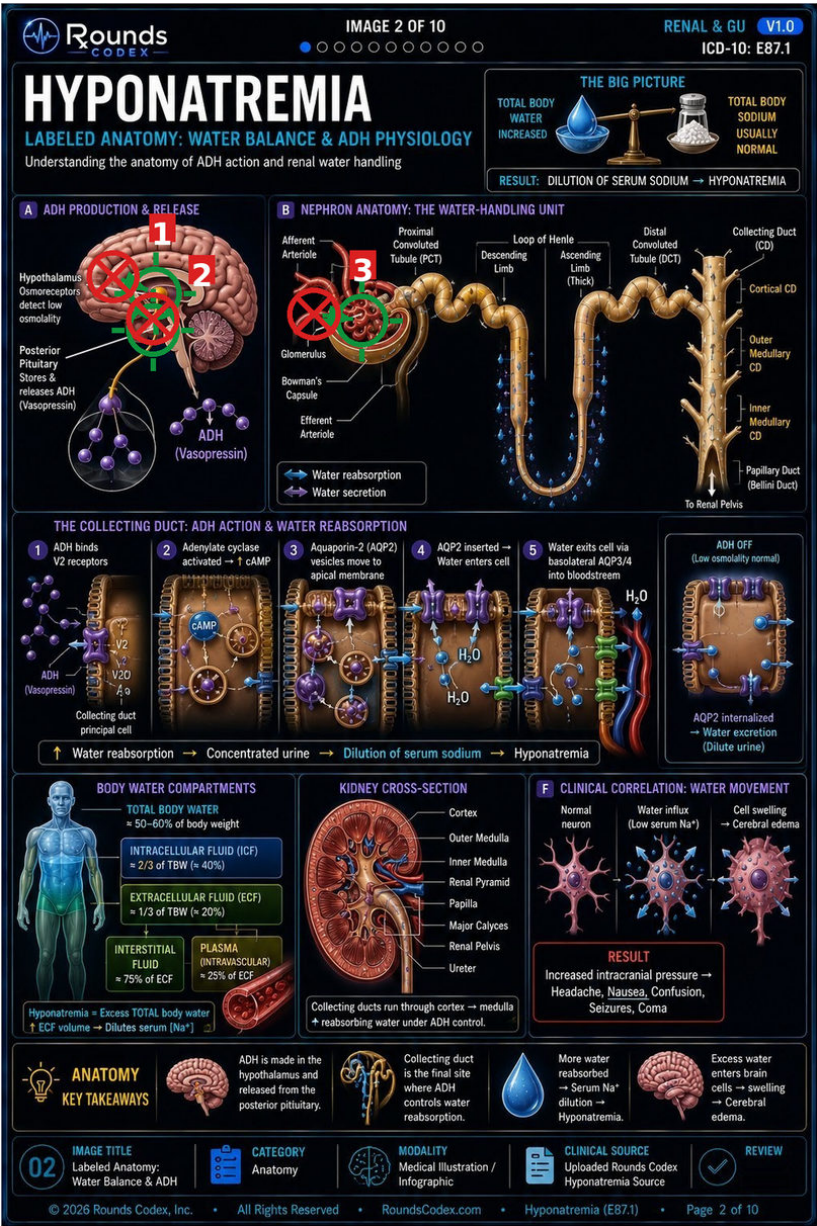
Corrections are integrated and deployed in batches of about six pages. Each one is measured against the sheet before it ships - what was fixed, what was missed, and what was left - and that measurement is reported back honestly rather than assumed.

hyponatremia — page 2

ADH production & release; nephron anatomy · hyponatremia-02.jpgg · 1024×1536 px · 3 findings

3 rows flagged — all 3 measured and all 3 confirmed. First of the described-only pages to be measured.

The work order gave no coordinate for any row on this page; every endpoint below was found by reading the artwork at 3-5x and confirmed by sampling the tissue under it with the leader's own ink excluded.

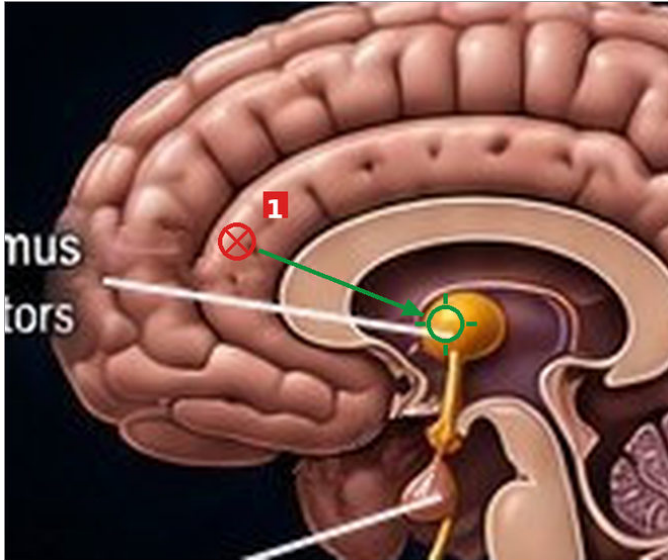


WHAT TO CHANGE

- Hypothalamus (osmoreceptors)**
 NOW (140,343) frontal-lobe cortical gyri, sampled (193,136,120). The leader is in TWO faint segments (ink peaks at x 105 and x 140) and both die in cortex
 MOVE TO (193,364) the gold hypothalamic nucleus at the top of the pituitary stalk, sampled (238,168,47) - unmistakable against anything around it
- Posterior Pituitary (stores & releases ADH) [CHECK]**
 NOW (190,396) sampled (118,60,32), dark suprasellar tissue 12 px ABOVE the gland - the order is right that it stops short
 MOVE TO (190,410) the pinkish-brown pituitary teardrop at the foot of the stalk, sampled (166,91,77)
- Glomerulus**
 NOW (392,390) sampled (150,68,58) - the red arteriole entering from the left, OUTSIDE Bowman's capsule
 MOVE TO (452,398) the bright red capillary tuft inside the capsule, sampled (184,85,68). Both arterioles are separately labelled on this panel, so the tuft is the only thing left unnamed

hyponatremia — page 2 — close-ups

red crossed circle = where it ends now green plain circle = where it belongs



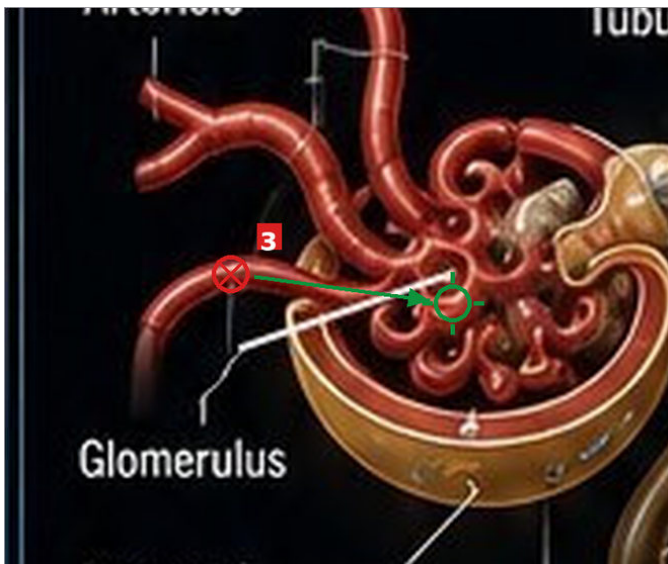
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hyponatremia — page 2 — notes

NOTES

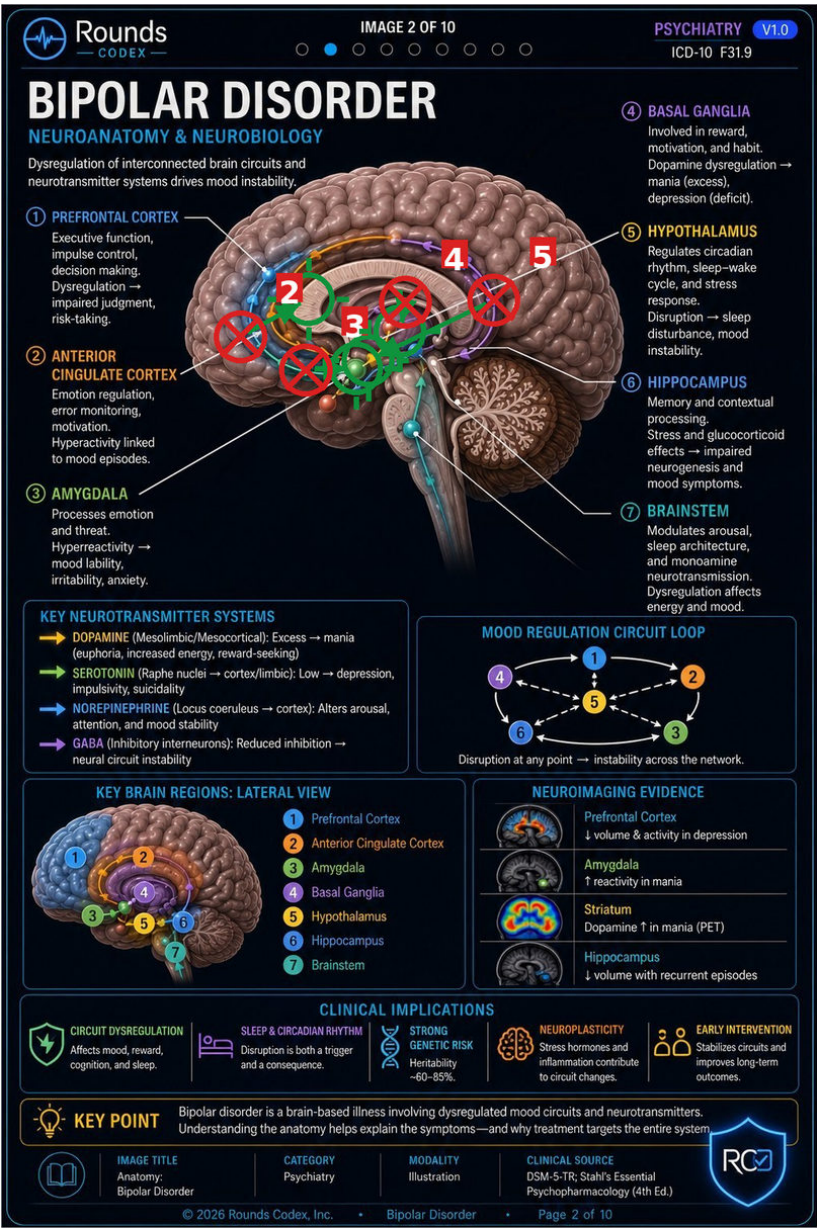
Bowman's Capsule on the same panel is CORRECT - its leader ends at (456,447) on the gold capsule ring, sampled (108,71,32). That matters for the fix: the Glomerulus leader has to reach past the capsule ring to the tuft inside it, so the two cannot simply be swapped end for end. ONE CORRECTION TO THE ORDER: it says the hypothalamic target is '~250px to the right'. Measured, it is 57 px (dx 53, dy 21). The direction and the target are right; the distance is not.

bipolar — page 2

Neuroanatomy & neurobiology — circuit figure · bipolar-02.jpg · 1024×1536 px · 4 findings

4 rows flagged — all 4 measured and all 4 confirmed, each endpoint carrying 37-56 px of leader ink.

A circuit diagram drawn over a midsagittal brain: coloured nodes joined by coloured pathway arcs, with white leaders running in from the numbered labels. Three of the four flagged leaders stop on a PATHWAY ARC rather than on the NODE the label names - the arcs are the conspicuous thing at that point on the page, and the leaders stop at the first one they meet.

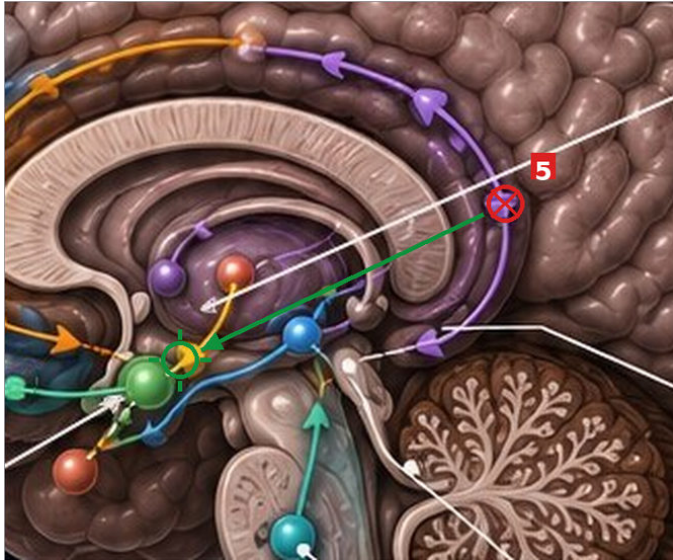


WHAT TO CHANGE

- 5 ⑤ **HYPOTHALAMUS**
NOW (618,371) sampled (123,99,127) - the purple GABA pathway arc, high and posterior in parietal white matter
MOVE TO (460,447) the gold circuit node at the base of the diencephalon below the thalamus, sampled (172,135,39)
- 2 ② **ANTERIOR CINGULATE CORTEX**
NOW (300,417) sampled (144,138,147) - the blue-tinted prefrontal band, well anterior to the cingulate
MOVE TO (385,370) the cingulate gyrus immediately above the genu of the corpus callosum, sampled (204,167,149), where the orange arc runs
- 3 ③ **AMYGDALA [CHECK]**
NOW (382,459) sampled (107,176,148) - the GREEN PATHWAY ARC, not the node
MOVE TO (445,457) the green amygdala sphere itself, sampled (118,186,112), 63 px to the right
- 4 ④ **BASAL GANGLIA [CHECK]**
NOW (507,374) sampled (96,77,77) - the pale corpus callosum at the upper margin of the deep-grey mass
MOVE TO (500,410) inside the purple basal ganglia / thalamic mass, sampled (109,73,88)

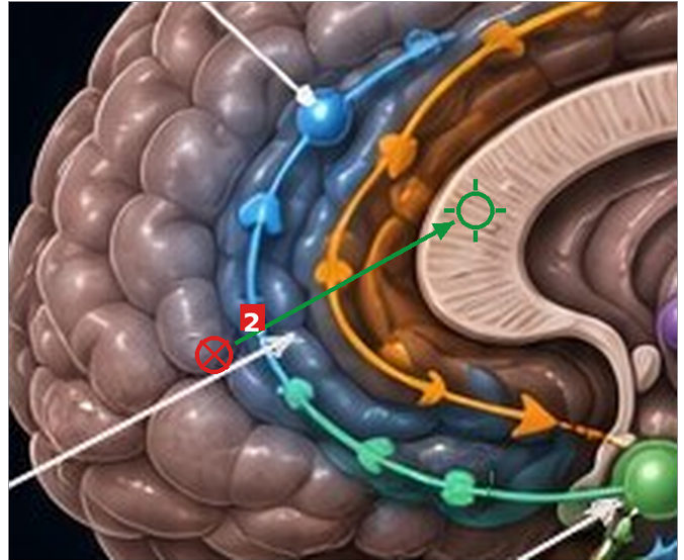
bipolar — page 2 — close-ups

red crossed circle = where it ends now green plain circle = where it belongs



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4 ④ BASAL GANGLIA [CHECK]

(507,374) sampled (96,77,77) - the pale corpus callosum at the upper margin of the deep-grey mass
-> (500,410) inside the purple basal ganglia / thalamic mass, sampled (109,73,88)

bipolar — page 2 — notes

NOTES

All four are short moves - 36 to 176 px - and none needs re-routing, because each label already approaches from the correct side. Row 3 is the cheapest and the clearest: 63 px along the leader's own axis, from the green arc onto the green sphere, both sampled and both unambiguous. ⑥ HIPPOCAMPUS and ⑦ BRAINSTEM were checked too and are not flagged here: ⑦ ends at (520,542) on the teal brainstem node, correct. ⑥'s leader ends at (617,523) on the CEREBELLUM, which is not the hippocampus - that row is NOT in the work order and needs the physician's read before we call it.

b12-anemia — page 2

Labeled anatomy: vitamin B12 & folate absorption · b12-anemia-02.jpg · 1024×1536 px · 4 findings

4 rows flagged — all 4 endpoints measured; 3 targets measured, 1 needs placing from source.

No coordinate was given for any row on this page; every endpoint below was read off the artwork and confirmed by sampling the tissue under it. Three of the four faults are the same shape: a badge or leader stops on the DUODENAL/BOWEL tube it happens to overlie rather than on the organ its legend names.

Rounds
CODEX

IMAGE 2 OF 10

HEME & ONC **V1.0**

ICD-10 D51.0

B12 / FOLATE DEFICIENCY ANEMIA

LABELED ANATOMY: VITAMIN B12 & FOLATE ABSORPTION

Two essential vitamins. Two different pathways.

VITAMIN B12 (COBALAMIN)

- Dietary B12**
From animal proteins (meat, fish, dairy, eggs).
- Stomach**
HCl releases B12 from food proteins.
B12 binds to R-proteins (haptocorrin).
- Parietal Cells**
Secrete intrinsic factor (IF) into the stomach.
- Duodenum**
Pancreatic proteases degrade R-proteins, freeing B12.
- Ileum (Terminal)**
B12 + IF bind to cubilin (IF receptor) on ileal enterocytes.
- Transport**
B12 enters portal blood bound to transcobalamin II and is delivered to tissues (liver stores large amounts).

FOLATE (FOLIC ACID)

- Dietary Folate**
From leafy greens, legumes, citrus, nuts, fortified grains.
- Small Intestine (Jejunum)**
Folate is released from food and absorbed primarily in the proximal small intestine by passive diffusion.
- Transport**
Folate enters portal blood as 5-methyltetrahydrofolate (5-MTHF) and is delivered to tissues.

KEY ANATOMY & FUNCTIONS

 Parietal cells produce intrinsic factor (IF). Essential for B12 absorption in terminal ileum.	 IF-B12 complex binds to cubilin receptors. Without IF or cubilin, B12 absorption is impaired.	 Liver stores ~2-5 mg of B12. Stores can last several years.	 Folate stores are small. Deficiency can develop in months.
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CLINICAL CORRELATION

 Loss of intrinsic factor (pernicious anemia) → impaired B12 absorption	 Ileal disease or resection (e.g., Crohn's) → impaired B12 absorption
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B12 ABSORPTION SUMMARY

FOLATE ABSORPTION SUMMARY

10

IMAGE TITLE

Labeled Anatomy:
B12 & Folate Absorption

CATEGORY

Hematology &
Oncology

MODALITY

Illustration

CLINICAL SOURCE

Hematology Guidelines • UpToDate •
Merck Manual

REVIEW

CLINICAL
PEARLS

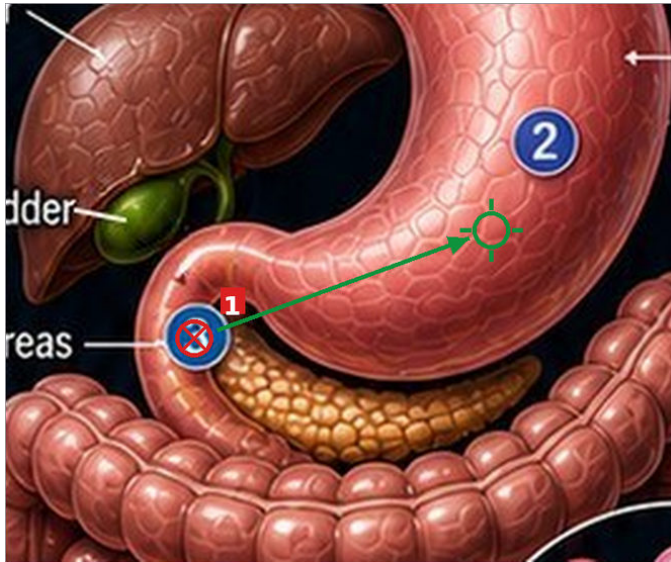
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WHAT TO CHANGE

- 1 ③ marker — legend "Parietal Cells"
NOW (442,513) the badge sits on the pink DUODENAL C-LOOP at the pylorus, sampled (225,132,117) just under it
MOVE TO (560,470) the gastric body/fundus where parietal cells actually sit, sampled (215,97,89)
- 2 **Pancreas [CHECK]**
NOW (431,516) the leader stops at the badge ③ ring on the pink duodenal wall - the orange pancreas does not begin until x=455, so it is ~24 px short
MOVE TO (490,535) the orange pancreas itself, sampled (196,120,65)
- 4 ⑥ marker — legend "Transport" (B12 enters portal blood) [CHECK]
NOW (499,873) sampled (112,52,48) - the pink BOWEL WALL. The leader runs left from the badge, curves up and dies on the gut
MOVE TO (450,875) the blue portal vein drawn immediately to its left, sampled (22,69,119) - unmistakable against the bowel
- 3 ⑤ Terminal ileum [CHECK]
NOW (502,829) the badge sits on the bowel loop descending on the patient's LEFT (the aorta/IVC pair run at x 430-467, so midline is ~455 and this loop is left of it)
MOVE TO the right lower quadrant, where the terminal ileum and cecum belong. PLACE FROM SOURCE - sampling the viewer-left lower quadrant returns background ((0,9,17) at (380,830)), so we could not find a distinctly drawn cecum to put a pixel on

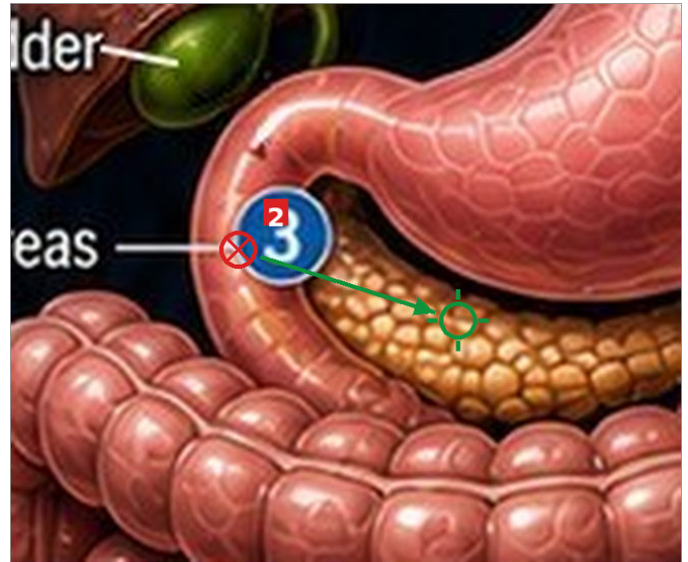
b12-anemia — page 2 — close-ups

red crossed circle = where it ends now green plain circle = where it belongs



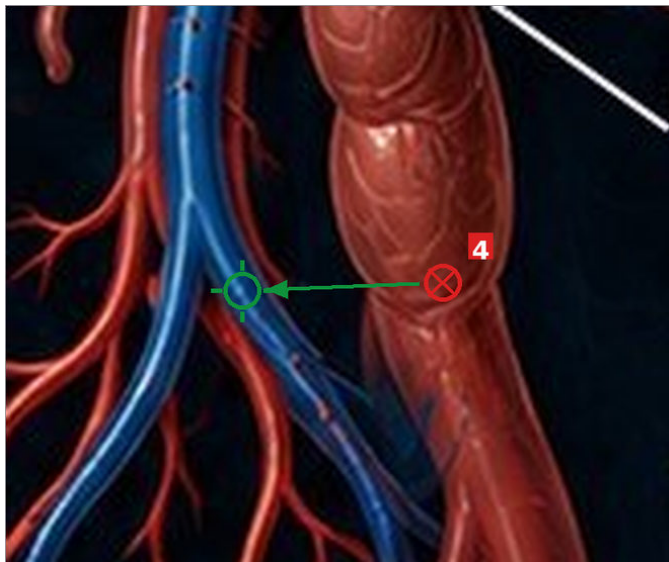
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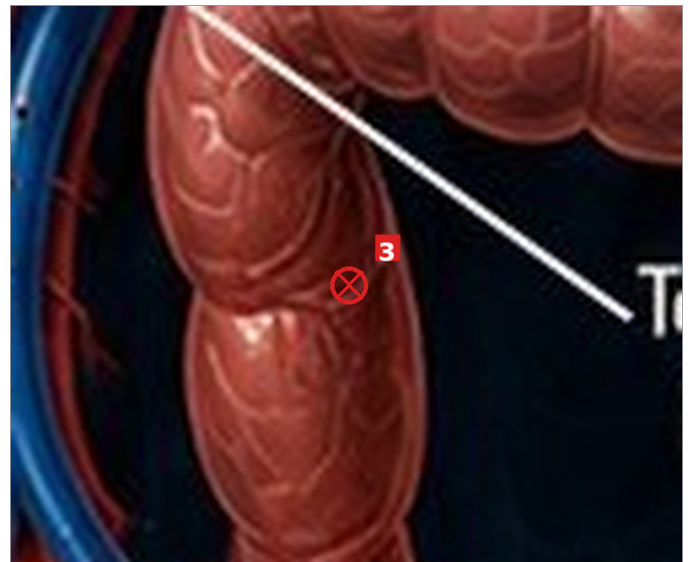
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b12-anemia — page 2 — notes

NOTES

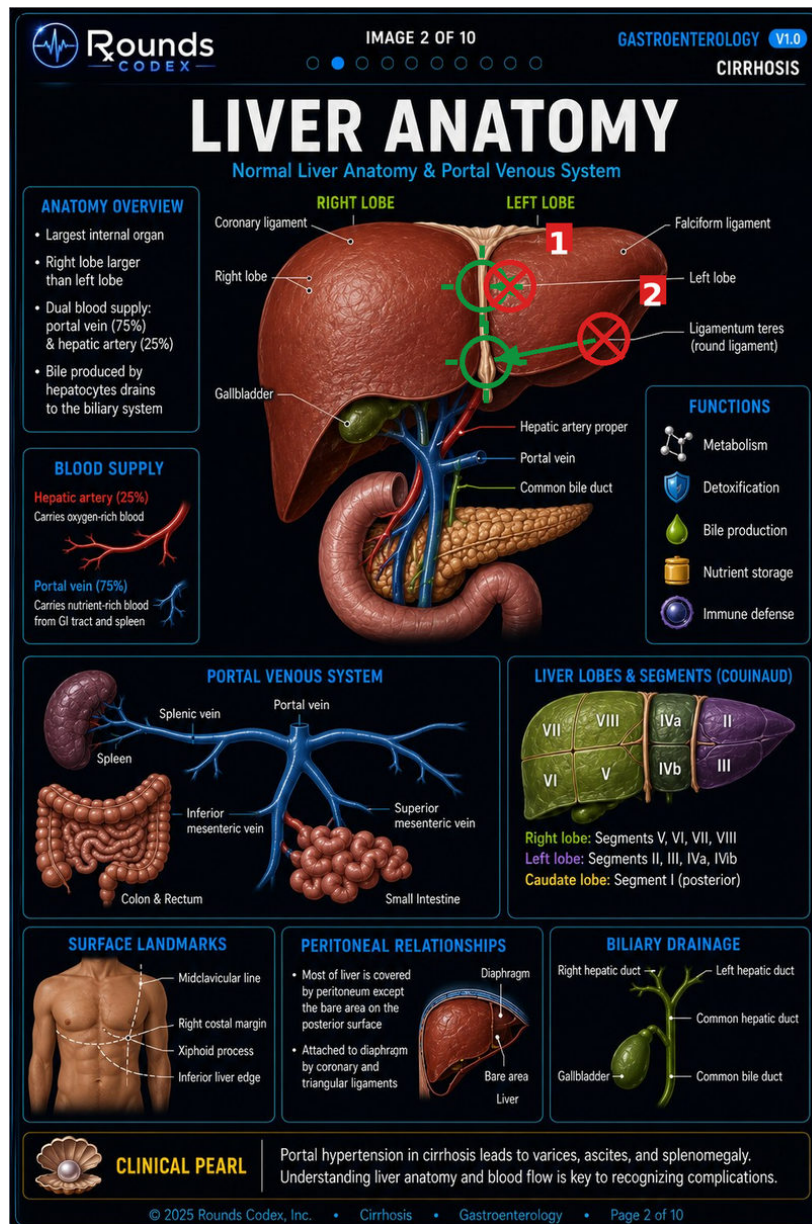
Row 1 has a second half the order flags and we confirm by absence: there is NO ④ marker anywhere on the figure, although the legend lists ④ Duodenum. So either the ③ badge is really the missing ④ sitting in the right place with the wrong number, or ④ was dropped. THAT CHANGES THE FIX - if it is a numbering error, renumber the badge rather than moving it, because it is already on the duodenum. Worth one look at the source file before anything is dragged. Rows 1, 2 and 4 are otherwise short, unambiguous moves.

cirrhosis — page 2

Normal liver anatomy & portal venous system · cirrhosis-02.jpg · 1024×1536 px · 2 findings

4 rows flagged — 2 measured and confirmed (main figure); rows 3 and 4 are on other panels and are NOT yet measured.

The main figure draws the falciform ligament as a narrow pale cream band. Measured at the leaders' own row it spans only x 590-597 - about 8 px - which is the whole difficulty: both flagged leaders overshoot it into liver parenchyma.



WHAT TO CHANGE

1 Falciform ligament

NOW (628,350) ink 24 confirms a real terminator; tissue (139,100,93), left-lobe parenchyma, 35 px lateral to the band
MOVE TO (593,350) the cream band itself, sampled (207,161,125) at x 590-597

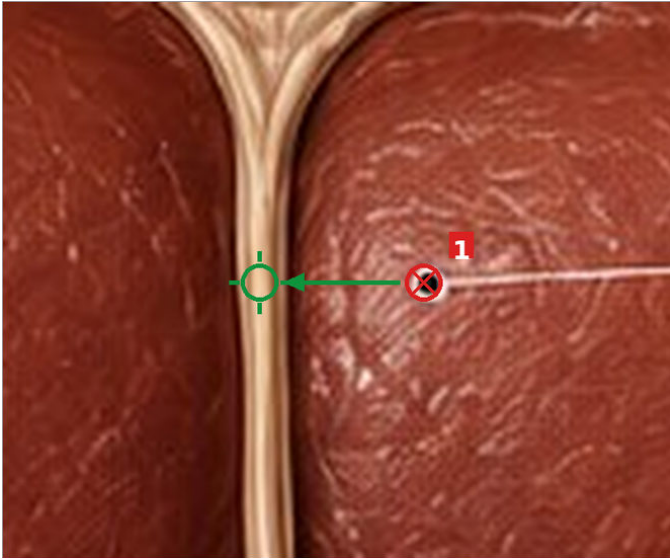
2 Ligamentum teres (round ligament)

NOW (746,416) ink 19; tissue (55,35,31), the shadowed inferolateral margin of the left lobe

MOVE TO (598,445) the cream cord descending below the liver from the falciform, sampled (199,155,122) through y 428-468

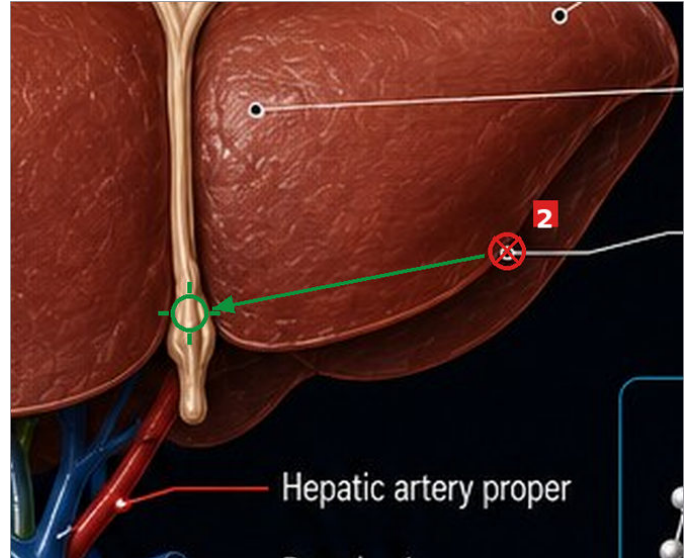
cirrhosis — page 2 — close-ups

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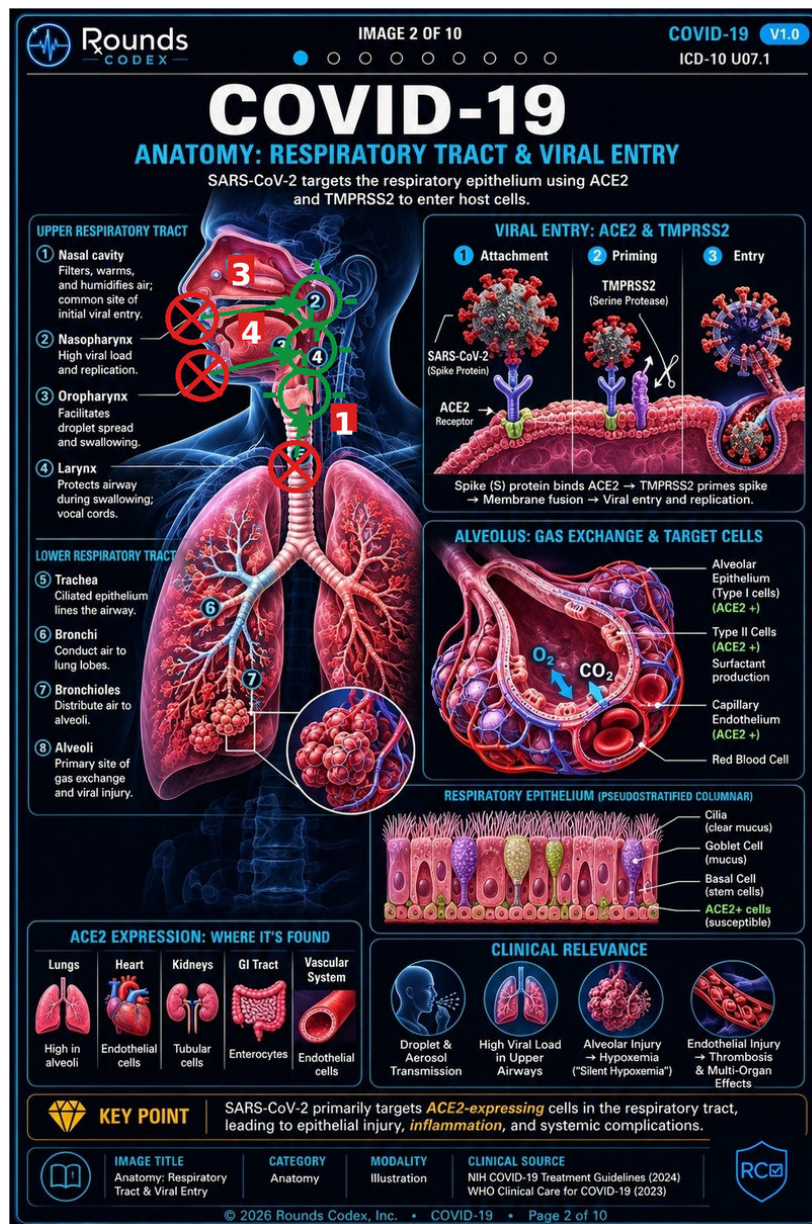
ONE CORRECTION TO THE ORDER'S MAGNITUDES, same as on hyponatremia: it puts row 1 at '~160px lateral to the midline' and row 2 at '~145px'. Measured, row 1 is 35 px from the band. The finding is right, the distance is not - do not drag by the order's numbers. ROWS 3 AND 4 ARE NOT MEASURED: row 3 Xiphoid process is in the SURFACE LANDMARKS panel and row 4 Diaphragm is in PERITONEAL RELATIONSHIPS. They are carried at the order's wording until the next pass.

covid — page 2

Respiratory tract & viral entry · covid-02.jpg · 1024×1536 px · 3 findings

4 rows flagged — 3 measured and confirmed (legend leaders on the main figure); row 2 is in the epithelium panel and is NOT yet measured.

Three of the four are LEGEND leaders - the lines running from the numbered list on the left across to the figure. Two of them never arrive: they stop in the black background outside the facial silhouette. The on-figure badges themselves are correctly placed, which is what makes this survive a skim - the reader sees a numbered badge in the right place and does not notice that the legend line beside it points at nothing.

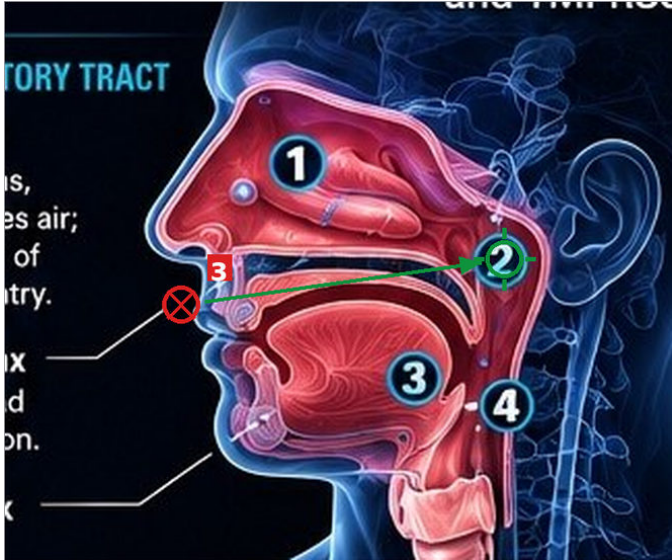


WHAT TO CHANGE

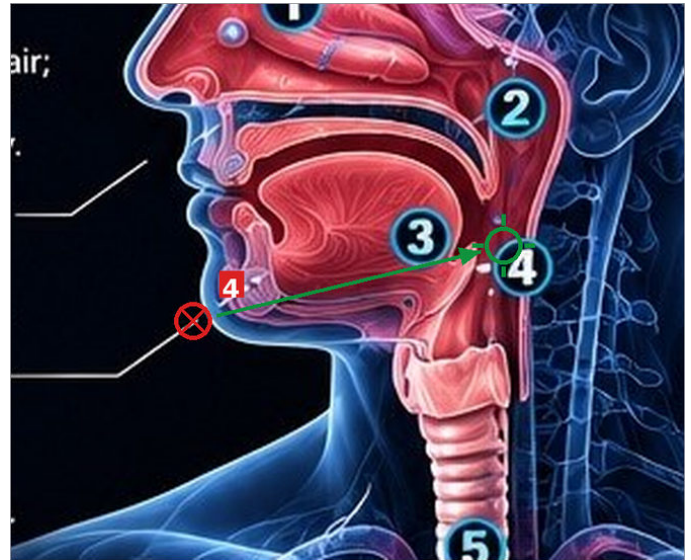
- 3 ② Nasopharynx (legend leader)
[CHECK]
NOW (229,389) sampled (2,4,10) - near-black EMPTY BACKGROUND just outside the facial silhouette, near the nose/upper lip
MOVE TO (385,367) the airway behind the nasal cavity, sampled (60,99,117), where the on-figure badge ② already sits correctly
- 4 ③ Oropharynx (legend leader)
[CHECK]
NOW (242,461) sampled (13,15,20) - near-black EMPTY BACKGROUND at the chin/submental outline
MOVE TO (378,428) the pharyngeal airway behind the tongue, sampled (71,65,93)
- 1 ④ Larynx (legend leader)
NOW (358,575) a horizontal leader at y≈575 running right across the whole neck and dying on the TRACHEA, just below the ⑤ Trachea badge at (365,560)
MOVE TO (370,485) the pale laryngeal cartilage below the epiglottis, sampled (242,165,166)

covid — page 2 — close-ups

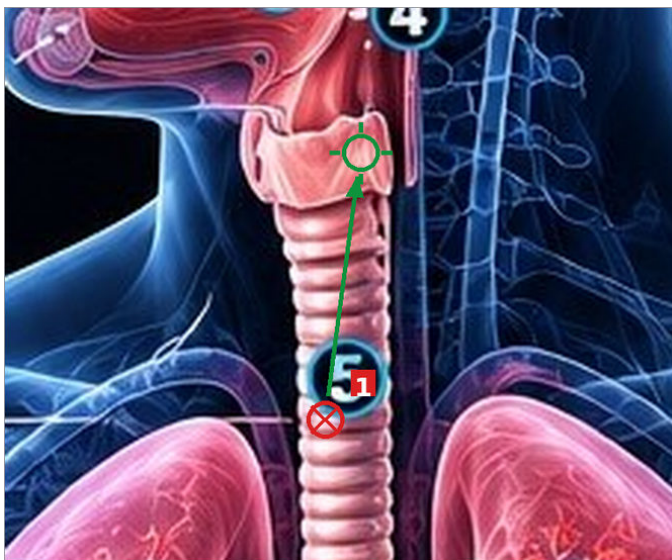
red crossed circle = where it ends now green plain circle = where it belongs



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covid — page 2 — notes

NOTES

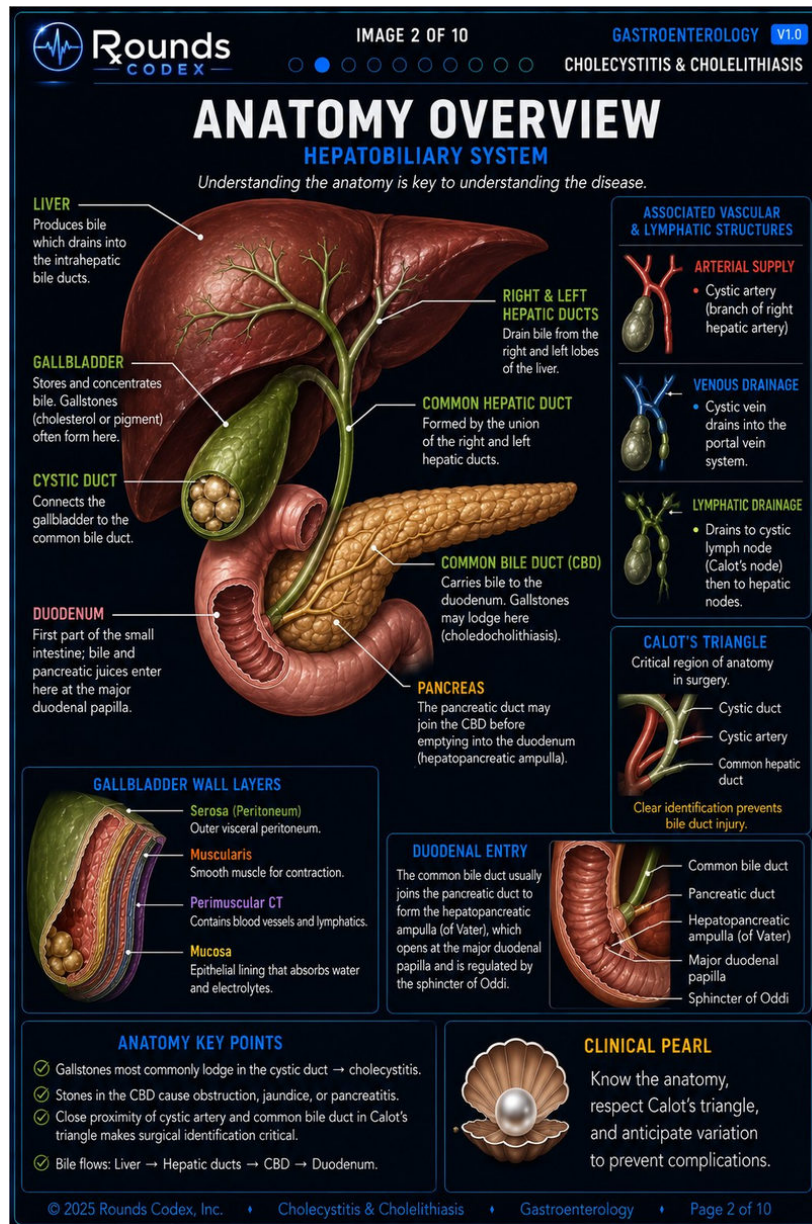
MEASURED CAVEAT on row 1: the ink run at y 575 is continuous from x 240 to about x 360, but the trachea's own pale rings also register there, so the terminus is given as $\approx(358,575)$ rather than to the pixel. The finding - that it ends on trachea rather than larynx - does not depend on that precision. ROW 2 NOT MEASURED: Basal Cell is in the RESPIRATORY EPITHELIUM panel (page x 478-990, y 990-1143); the order has it on the stalk of the purple goblet cell when it belongs on the squat cuboidal row at the basement membrane. Carried at the order's wording. PANELS SWEEPED: the main figure's badges ①-⑧ were checked and all sit on their structures - it is only the legend LINES that fail.

cholecystitis-p2

Anatomy Overview: Hepatobiliary System · cholecystitis-02.jpg · 1024×1536 px · 7 findings

NOT MEASURED BY US — 7 of 20 labels flagged — 5 to move · 2 to check

The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.



WHAT TO CHANGE

- CYSTIC DUCT (main figure)**
the gallstones inside the cut-open gallbladder fundus
MOVE TO the green tube arising from the gallbladder neck that arches right to join the hepatic duct
- COMMON BILE DUCT (CBD) (main figure)**
pancreas parenchyma immediately above the yellow pancreatic duct
MOVE TO the green duct descending behind the pancreatic head
- COMMON HEPATIC DUCT (main figure)**
the bile duct ~45px BELOW the cystic-duct junction, i.e. the common bile duct
MOVE TO the short segment between the R+L hepatic confluence and the cystic-duct junction
- Mucosa (Gallbladder wall layers panel)**
the outermost blue/purple band of the layered fan
MOVE TO the red folded inner lining facing the lumen and the gallstones, ~30-60px further in
- Sphincter of Oddi (Duodenal entry panel)**
the outer duodenal wall at the bottom of the C-loop
MOVE TO the muscle at the ampulla/papilla, higher and to the left
- Muscularis and "Perimuscular CT" (Gallbladder wall layers panel) [CHECK]**
land on: the same outer blue/purple boundary at different heights along the curve
MOVE TO two distinct bands within the fan
- Cystic artery (Calot's triangle panel) [CHECK]**
the green bile duct at the point where the red artery passes behind it
MOVE TO on the red artery itself, which the leader passes over on its way in

NOTES

The order's FINDINGS have held up on every page we checked. Its DISTANCES have not: '~250px' measured 57 px on hyponatremia and '~160px' measured 35 px on cirrhosis. Work from the description and the source file, not from the order's numbers.

croup-p2

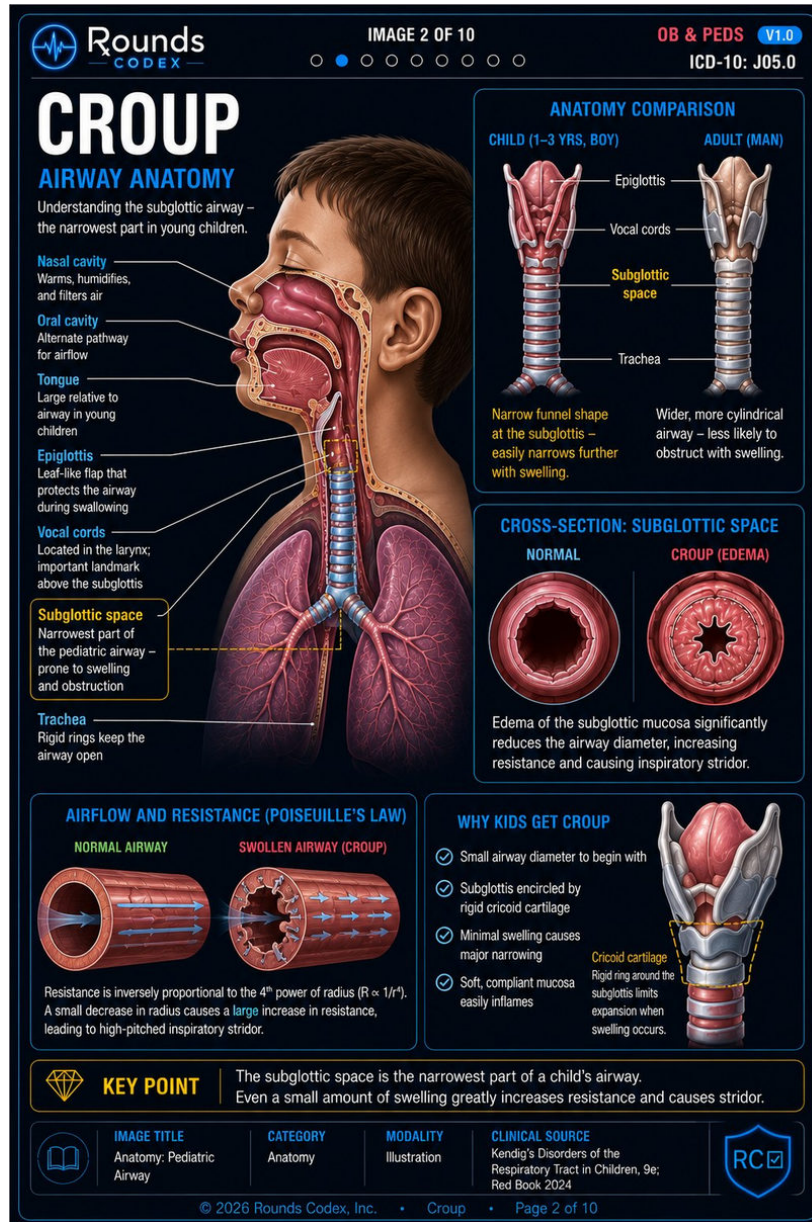
Anatomy: Pediatric Airway · croup-02.jpg · 1024×1536 px · 4 findings

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The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.

WHAT TO CHANGE

- 1 Trachea (main sagittal figure)**
left lung parenchyma at the medial border, well BELOW the carina
MOVE TO the ringed blue-grey tube in the neck between larynx and carina
- 2 Vocal cords (main sagittal figure)**
[CHECK]
a point inside the yellow dashed "Subglottic space" callout box
MOVE TO just ABOVE the subglottis, at the glottic level
- 3 Epiglottis (main sagittal figure)**
[CHECK]
soft tissue in the laryngeal vestibule roughly 15px posterior to the pale leaf-shaped epiglottis; the line crosses the epiglottis and overshoots it
MOVE TO on the epiglottis itself
- 4 Oral cavity (main sagittal figure)**
[CHECK]
the lips at the mouth opening, then resumes as short stubs with dots on the body of the tongue
MOVE TO the airspace of the oral cavity above the tongue



NOTES

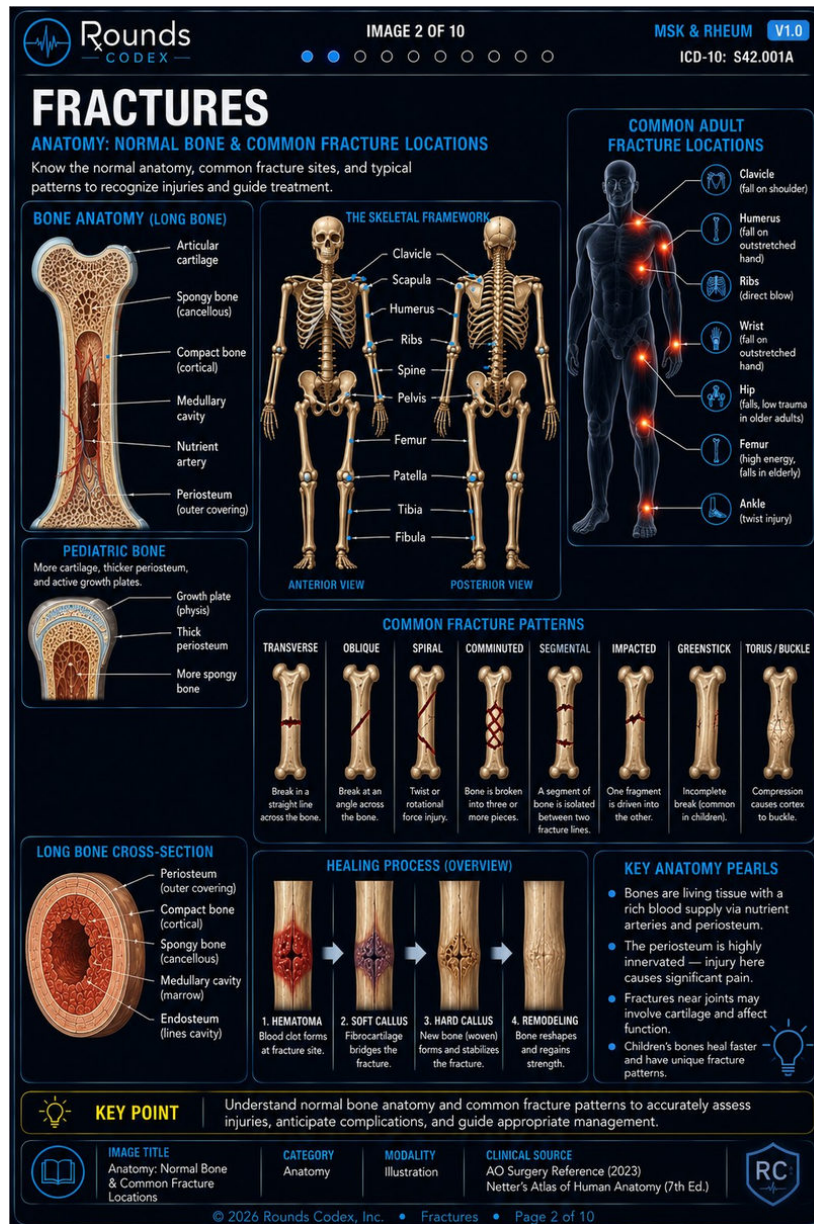
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fracture-p2

Anatomy: Normal Bone & Common Fracture Locations · fracture-02.jpg · 1024×1536 px · 7 findings

NOT MEASURED BY US — 7 of 31 labels flagged — 4 to move · 3 to check

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WHAT TO CHANGE

- 1 Ribs (THE SKELETAL FRAMEWORK panel)**
the ELBOW marker (distal humerus/olecranon) on BOTH the anterior and the posterior skeleton
MOVE TO the rib cage, which is drawn plainly to the left/right of the arrow and even carries its own unlabelled marker on the posterior view
- 2 Spine (THE SKELETAL FRAMEWORK panel)**
the FOREARM (proximal radius/ulna) marker on the anterior skeleton, and on empty space over the posterior forearm on the right
MOVE TO the vertebral column, which has its own unlabelled marker beside the lumbar spine
- 3 Tibia (THE SKELETAL FRAMEWORK panel)**
the thin LATERAL bone (fibula) on both skeletons - the same bone the "Fibula" arrow 20px lower points at
MOVE TO the thick medial bone of the leg
- 4 Growth plate (physis) (PEDIATRIC BONE panel)**
the grey outer covering at the metaphyseal shoulder, i.e. the same periosteal band that the "Thick periosteum" arrow below it points at
MOVE TO the pale blue physeal band arcing across the bone, ~30px inferomedial
- 5 Scapula (THE SKELETAL FRAMEWORK panel) [CHECK]**
the glenohumeral/humeral-head marker on both views; on the posterior view the scapular body is drawn immediately lateral to it and carries its own unlabelled marker
MOVE TO the scapular blade. Arguable, since the glenoid is part of the scapula
- 6 Nutrient artery (BONE ANATOMY long-bone panel) [CHECK]**
the dark red medullary marrow, ~12px right of the red arterial tree and the grey vessel it presumably names
MOVE TO the red branching vessel in the marrow cavity
- 7 Medullary cavity (marrow) (LONG BONE CROSS-SECTION panel) [CHECK]**
the red cancellous ring, short of the dark swirled central marrow; no leader in that panel reaches the marrow cavity
MOVE TO the dark central region

NOTES

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gi-bleed-p2

Upper GI Anatomy: Structures at Risk for Bleeding · gi-bleed-02.jpg · 1024×1536 px · 4 findings

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Roundscodex

IMAGE 2 OF 10

GASTROENTEROLOGY V1.0

Upper GI Bleed (ICD-10 K92.2)

UPPER GI ANATOMY

STRUCTURES AT RISK FOR BLEEDING

Upper GI bleeding arises from structures proximal to the ligament of Treitz.

THE LIGAMENT OF TREITZ

A fibromuscular band anchoring the duodenojejunal junction to the right crus of the diaphragm.
All bleeding sources proximal to this point are considered **UPPER GI**.

ARTERIAL SUPPLY (AT RISK)

Esophagus

- Inferior thyroid aa.
- Bronchial aa.
- Esophageal aa.
- Left gastric a.

Stomach

- Left gastric a.
- Right gastric a.
- Left gastroepiploic a.
- Right gastroepiploic a.
- Short gastric aa.

Duodenum

- Gastroduodenal a.
- Superior pancreaticoduodenal a.
- Posterior superior pancreaticoduodenal a.

WALL ANATOMY (FROM LUMEN OUTWARD)

- Mucosa**
Epithelium, lamina propria, muscularis mucosae
Site of most ulcers and erosions
- Submucosa**
Vessels, lymphatics, Meissner plexus
Varices can extend into submucosa
- Muscularis Externa**
Inner circular & outer longitudinal layers
Aids peristalsis and mixing
- Serosa / Adventitia**
Serosal covering (stomach, duodenum)
Adventitia where retroperitoneal

KEY ANATOMIC RELATIONSHIPS

- Esophagus**
Posterior to trachea; passes through diaphragm at T10.
- Stomach**
Under left hemidiaphragm; anterior to pancreas and transverse colon.
- Duodenum**
"C"-shaped; wraps around pancreas head; largely retroperitoneal.
- Liver**
Receives portal blood from GI tract; key in variceal formation.
- Portal System**
Drains venous blood from GI tract to liver — source of varices.

CLINICAL PEARLS

- ✓ Peptic ulcers most commonly occur in the antrum and duodenal bulb.
- ✓ Varices form in distal esophagus and gastric fundus along the lesser curvature.
- ✓ Dieulafoy lesion classically arises on the lesser curvature of the stomach within 6 cm of the GEJ.
- ✓ The duodenum is a common site for ulcers and vascular lesions.

MODULE TITLE

Upper GI Bleed Foundations

CATEGORY

Gastroenterology

DIFFICULTY

Intermediate

CLINICAL IMPACT

High

REVIEW

Anatomy & Physiology

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WHAT TO CHANGE

1 Cardia

the outer wall of the fundus dome, top-right of the stomach, well above and right of the esophagogastric junction
MOVE TO the EG junction where the esophagus joins the stomach (upper left of the stomach)

2 Fundus [CHECK]

greater-curvature wall at/just below the level of the cardia, i.e. upper body
MOVE TO the dome above the cardia (which is where the "Cardia" dot sits)

3 Antrum [CHECK]

mucosal surface of the greater curvature at the lower body, far from the pylorus
MOVE TO the distal stomach adjacent to the pylorus, which in this drawing is upper/left where the duodenal bulb begins

4 Serosa / Adventitia [CHECK]

the outer longitudinal muscle band of the muscularis externa on the block's right face, ~1 layer above the outermost pale layer
MOVE TO the outermost pale layer below the subserosal (blue) vessel

Rounds Codex — mislabelled anatomy corrections

gi-bleed-p2

21

NOTES

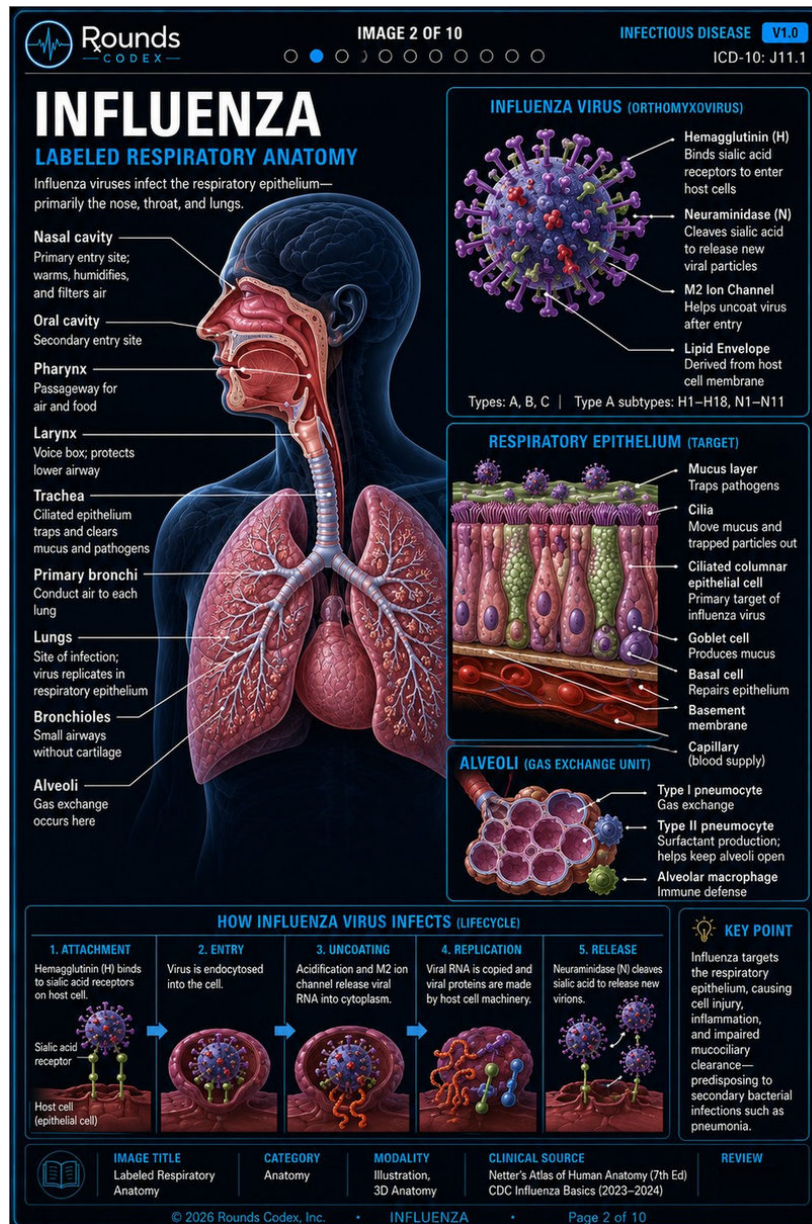
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influenza-p2

Labeled Respiratory Anatomy · influenza-02.jpg · 1024×1536 px · 6 findings

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WHAT TO CHANGE

- 1 Oral cavity**
the nostril / nasal vestibule at the tip of the nose
MOVE TO the mouth cavity below the hard palate (the Pharynx leader already reaches past it to the oropharynx)
- 2 Neuraminidase (N)**
black space immediately above a PURPLE spike, the same spike type "Hemagglutinin (H)" points at; the distinct olive-green spikes are left unlabelled
MOVE TO the olive-green spike type
- 3 Goblet cell**
the purple nucleus inside a pink ciliated columnar cell (the cell type labelled one line above)
MOVE TO one of the green mucus-filled cells
- 4 Primary bronchi [CHECK]**
a segmental bronchus deep inside the right lung field, about two generations distal to the main bronchus
MOVE TO the large ringed tube just off the carina
- 5 M2 Ion Channel [CHECK]**
the envelope margin among purple spikes, with no distinct channel structure at the endpoint
MOVE TO one of the small red/dark membrane structures
- 6 Basement membrane [CHECK]**
a second leader from this label ends in the capillary lumen ~12px below the tan membrane band
MOVE TO only the dogleg leader (which does land correctly on the membrane) should be there

NOTES

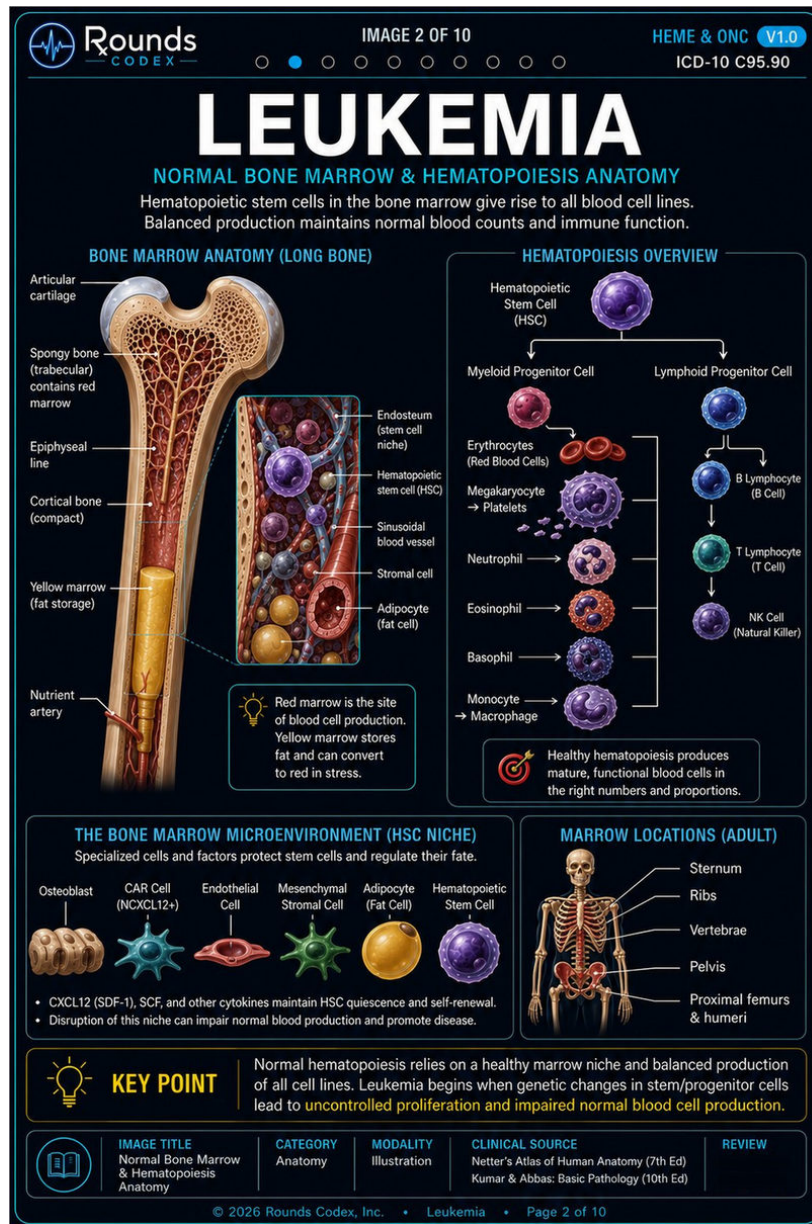
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leukemia-p2

Normal Bone Marrow & Hematopoiesis Anatomy · leukemia-02.jpg · 1024×1536 px · 6 findings

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WHAT TO CHANGE

- Sternum**
the humeral head / shoulder joint on the figure's left
MOVE TO the pale vertical midline bone in the front of the chest
- Vertebrae**
a lower rib on the figure's left, ~30px lateral to the spine
MOVE TO the red vertebral column in the midline
- Endosteum (stem cell niche)**
a blue-grey sinusoidal vessel on the far right of the inset, ~100px from the bone
MOVE TO the marrow-facing surface of the tan bone at the left edge of the inset
- Adipocyte (fat cell)**
the lumen of the large red blood vessel (red cells visible inside it). A second, orphan line segment does end on a golden fat cell but connects to no label
MOVE TO one of the golden fat spheres
- Epiphyseal line**
red marrow just inside the cortex, in the mid-diaphysis, ~90px below the head/neck junction
MOVE TO the growth-plate line at the epiphysis/metaphysis junction (none is clearly drawn)
- Hematopoietic stem cell (HSC)**
[CHECK]
a small beige/cream cell
MOVE TO probably the large lavender cell 100px to its left, which matches the purple HSC colour used in the Hematopoiesis and Niche panels on the same page

NOTES

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lung-cancer-p2

Labeled Anatomy & Typical Tumor Locations · lung-cancer-02.jpg · 800×1200 px · 7 findings

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IMAGE 2 OF 10 RESPIRATORY V1.0 ICD-10: C34.90

LUNG CANCER

Labeled Anatomy & Typical Tumor Locations

Lung cancers arise from bronchial or alveolar epithelium. Tumor location often correlates with histologic type, risk factors, and clinical syndromes.

BRONCHOPULMONARY ANATOMY

Trachea
Right main bronchus
Bronchial epithelium (origin of most cancers)
Segmental bronchi
Alveolar epithelium (origin of adenocarcinoma)
Alveoli
Left main bronchus
Lobar bronchi
Pulmonary artery (to lungs)
Pulmonary vein (from lungs)

TYPICAL TUMOR LOCATIONS

CENTRAL (HILAR) TUMORS

- Arise near the hilum and main bronchi
- Classically Squamous Cell Carcinoma or Small Cell Lung Cancer
- Strongly associated with smoking

PERIPHERAL (PARENCHYMAL) TUMORS

- Arise in distal bronchioles or alveoli
- Most commonly Adenocarcinoma
- More common in non-smokers

APICAL (SUPERIOR SULCUS) TUMOR (Pancoast Tumor)

Brachial plexus
Subclavian artery
Subclavian vein
1st rib
Superior sulcus (Pancoast) tumor

Can invade and cause:
• Shoulder/arm pain
• Horner syndrome (ptosis, miosis, anhidrosis)
• Upper extremity weakness

LYMPH NODE STATIONS (N COMMON LOCATIONS)

NODAL STATIONS

- 1 Supraclavicular
- 2 Upper paratracheal
- 3 Lower paratracheal
- 4 Subcarinal
- 5 Hilar (interlobar)
- 6 Ipsilateral pulmonary ligament

EBUS (endobronchial ultrasound) is commonly used for mediastinal nodal staging and sampling.

SVC SYNDROME

- Central tumor compresses the SVC
- Facial/neck swelling
- Distended neck veins
- Dyspnea, headache

HISTOLOGY CORRELATION WITH LOCATION

HISTOLOGIC TYPE	TYPICAL LOCATION	KEY POINTS	DRIVER MUTATIONS
Adenocarcinoma	Peripheral (subpleural)	• Most common subtype overall • Most common in non-smokers • Often associated with EGFR, ALK, ROS1, KRAS mutations	EGFR, ALK, ROS1, KRAS G12C
Squamous Cell Carcinoma	Central (hilar/main bronchi)	• Strongly associated with smoking • Linked to paraneoplastic syndromes • May cavitate	Less common driver mutations
Small Cell Lung Cancer	Central (hilar/main bronchi)	• Neuroendocrine tumor • Rapid growth and early metastasis • Strongly associated with paraneoplastic syndromes (SIADH, Cushing)	Rare targetable mutations
Large Cell Carcinoma	Peripheral or Central	• Poorly differentiated NSCLC • Diagnosis of exclusion	Variable
Driver Mutations	Primarily Adenocarcinoma	• Molecular profiling is essential to guide targeted therapy • PD-L1 testing guides immunotherapy	EGFR, ALK, ROS1, KRAS G12C, PD-L1

CLINICAL PEARLS

- Smoking-related cancers are usually central; non-smokers often have peripheral adenocarcinoma.
- Central tumors are more likely to cause obstructive symptoms and paraneoplastic syndromes.
- Location helps predict histology and guides the diagnostic approach and biopsy strategy.
- PD-L1 testing is essential before selecting immunotherapy in advanced NSCLC.
- Molecular profiling (EGFR, ALK, ROS1, KRAS G12C, etc.) is essential for all advanced or recurrent NSCLC.

KEY TAKEAWAY

Tumor location predicts histology. Peripheral tumors are usually adenocarcinoma, whereas central tumors are typically squamous cell carcinoma or small cell lung cancer. Histology, staging, and molecular profiling guide modern treatment.

IMAGE TITLE: Labeled Anatomy & Typical Tumor Locations
CATEGORY: Respiratory
DIFFICULTY: Foundational
CLINICAL SOURCE: UpToDate, NCCN Guidelines, ACCP Guidelines
REVIEW: HIGH-YIELD CLINICAL

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WHAT TO CHANGE

- 1 Right main bronchus**
a fine peripheral bronchus high in the right upper lobe; the ringed main bronchus at that x is ~80px lower
MOVE TO the large ringed tube running down-left from the carina
- 2 Left main bronchus**
a fine peripheral bronchus high in the left upper lobe, mirror-image of the right
MOVE TO the large ringed tube running down-right from the carina
- 3 Lobar bronchi [CHECK]**
a small branch in the upper left lung, distal to the lobar divisions
MOVE TO — see appendix
- 4 Segmental bronchi [CHECK]**
the fine parenchymal network ~20px lateral to the nearest ringed segmental bronchus
MOVE TO — see appendix
- 5 Pulmonary artery (to lungs) [CHECK]**
a pale-pink peripheral vessel of the parenchymal network; the dark navy vessels (which match this label's blue bullet) run ~13px to its left
MOVE TO — see appendix
- 6 Brachial plexus / "Subclavian artery" [CHECK]**
the two leaders converge on ONE shared orange arrowhead, so at most one of them can be pointing at its own structure; the arrowhead lands on the red band above the subclavian vein rather than on the cream nerve cords just left of it
MOVE TO — see appendix
- 7 1st rib [CHECK]**
arrow tip lands on the pale strand / vessel bundle running vertically behind the subclavian vessels; no rib is resolvable at the endpoint (the ribs are the tan arcs above and to the right). Panel is small and 800px wide - low confidence
MOVE TO — see appendix

NOTES

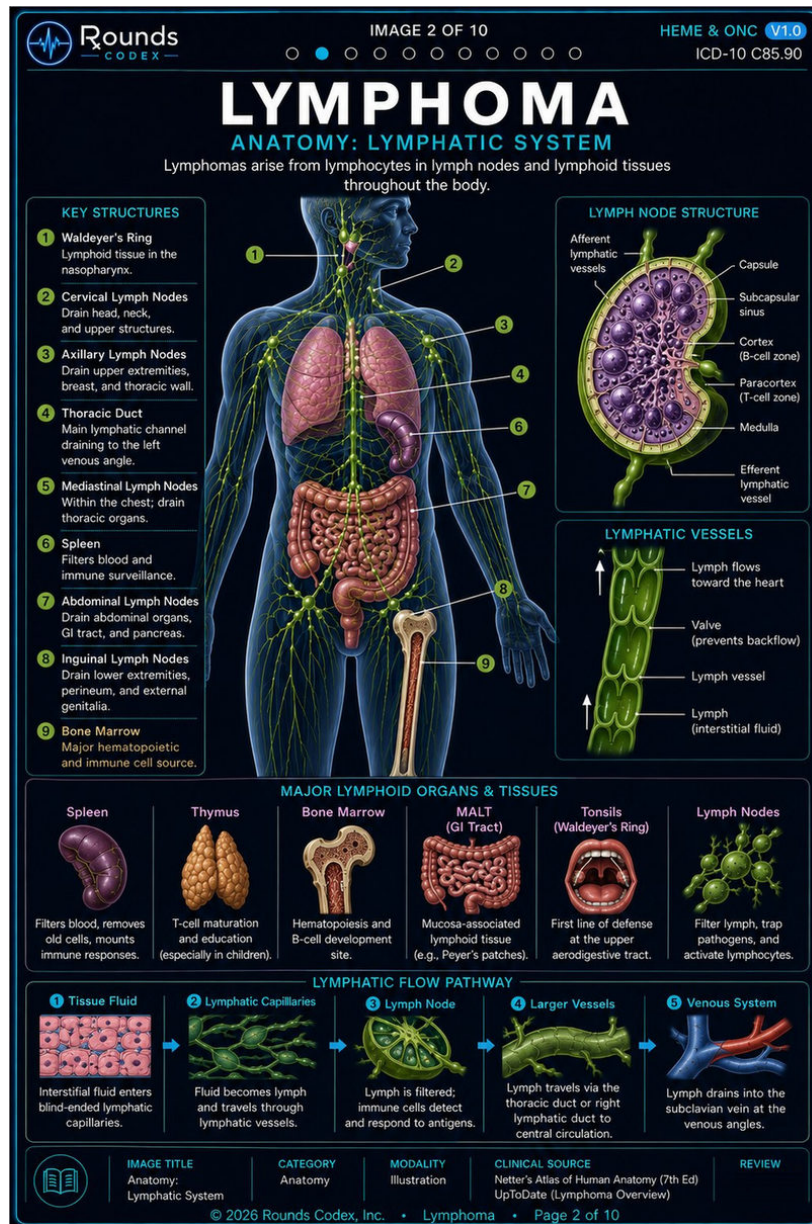
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lymphoma-p2

Anatomy: Lymphatic System · lymphoma-02.jpg · 1024×1536 px · 6 findings

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WHAT TO CHANGE

- 5 Mediastinal Lymph Nodes**
nothing - there is no numbered 5 marker or leader anywhere on the body figure (the right-hand column runs 2,3,4,6,7,8,9)
MOVE TO a marker on the paratracheal/hilar nodes between the lungs
- 8 Inguinal Lymph Nodes**
the head/neck of the cut femur (pale bone), the same bone that marker 9 labels
MOVE TO the large green inguinal node cluster ~30px to its left in the groin
- 7 Abdominal Lymph Nodes [CHECK]**
the wall of the descending colon (a haustrum), not on a node
MOVE TO one of the green mesenteric/para-aortic nodes drawn just medial to it
- Capsule [CHECK]**
the pale yellow-green band, i.e. the same band the "Subcapsular sinus" leader ends in
MOVE TO the outer dark-green rim
- Paracortex (T-cell zone)**
the outer dark-green capsule, outside the node parenchyma altogether
MOVE TO the pink-purple interfollicular zone between the follicles and the medulla
- Medulla**
the pale yellow-green subcapsular band at the node's outer edge
MOVE TO the central pink medullary cords/sinuses

NOTES

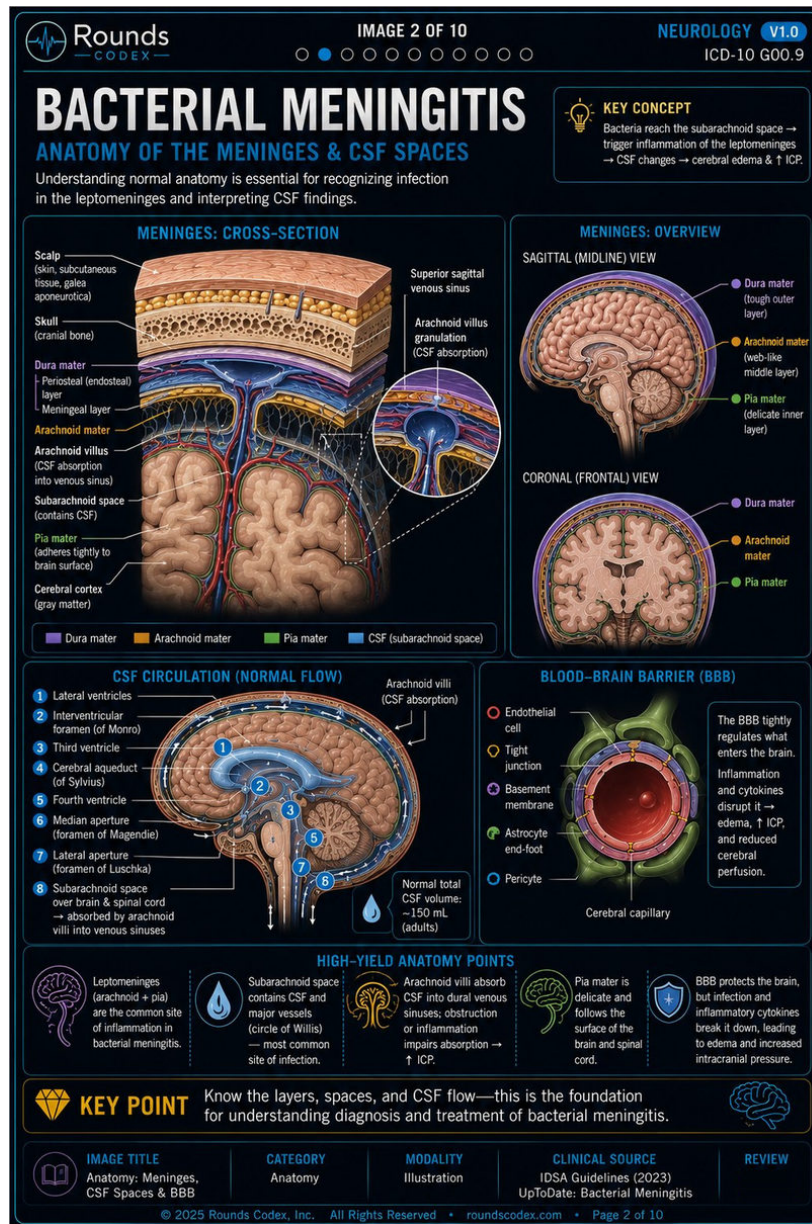
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meningitis-p2

Anatomy of the Meninges & CSF Spaces · meningitis-02.jpg · 1024×1536 px · 12 findings

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The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.



WHAT TO CHANGE

- Meningeal layer (dura sub-label)**
the golden/orange arachnoid mater band
MOVE TO the deeper of the two purple/blue dural layers, above the arachnoid
- Arachnoid villus (CSF absorption into venous sinus)**
the pial vessel / cortical surface at the top-left of the brain in the main cross-section
MOVE TO the villus protruding into the sagittal sinus (correctly shown in the inset, which the separate "Arachnoid villus granulation" label does point at)
- Pia mater (cross-section) [CHECK]**
a sulcus ~30px deep inside the gyral mass
MOVE TO the thin green line on the brain surface
- Superior sagittal venous sinus [CHECK]**
the purple dura at the top-left of the inset circle (the long bracket is hard to follow)
MOVE TO the blue sinus lumen in the middle of the inset
- Pia mater (sagittal overview) [CHECK]**
the tan arachnoid/dural band, about one layer outside the green pial line
MOVE TO the green line on the cerebellar/occipital surface
- Dura mater (coronal overview) [CHECK]**
the arachnoid/subarachnoid zone about one layer inside the purple band
MOVE TO the purple dura
- Pia mater (coronal overview) [CHECK]**
the orange arachnoid band
MOVE TO the green pial line on the cortical surface
- 4 Cerebral aqueduct (of Sylvius)**
the anterior horn of the lateral ventricle in the frontal lobe, and there is no 4 marker anywhere on the figure
MOVE TO the narrow channel between the third and fourth ventricles
- 6 Median aperture (foramen of Magendie)**
the inferior occipital cortex / inner skull base, and there is no 6 marker anywhere on the figure
MOVE TO the outlet at the inferior fourth ventricle
- 3 Third ventricle [CHECK]**
the body of the lateral ventricle (near the 1 marker)
MOVE TO the midline slit below the foramen of Monro - note a correctly placed 3 circle does exist on the figure

NOTES

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metabolic-syndrome-p1

Normal Anatomy & Visceral Adiposity · metabolic-syndrome-01.jpg · 800×1200 px · 7 findings

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01 IMAGE TITLE: Normal Anatomy & Visceral Adiposity CATEGORY: Pathophysiology DIFFICULTY: Foundation CLINICAL SOURCE: AMA/ACC, IDF, NHLBI REVIEW: High Yield

THE MEDICAL STUDENT COMPANION

IMAGE 1 OF 10

ENDOCRINOLOGY V1.0 ICD-10: E88.81

METABOLIC SYNDROME

NORMAL ANATOMY & VISCERAL ADIPOSITY

A cluster of cardiometabolic risk factors driven by excess visceral fat and insulin resistance, leading to dyslipidemia, hypertension, and dysglycemia.

NORMAL ANATOMY

Balanced adipose distribution supports healthy metabolic function.

Subcutaneous Adipose Tissue (SAT)
Visceral Adipose Tissue (VAT)
Abdominal Musculature
Peritoneal Cavity

In normal anatomy, visceral fat is limited and metabolically healthy, with low inflammatory activity and normal insulin sensitivity.

SUBCUTANEOUS vs VISCERAL FAT

LOW VISCERAL FAT (HEALTHIER PROFILE) HIGH VISCERAL FAT (METABOLIC RISK)

- More subcutaneous fat
- Less visceral fat
- Lower cardiometabolic risk
- Better insulin sensitivity

- Excess visceral fat
- Greater inflammation
- Insulin resistance
- Higher CVD & T2DM risk

VISCERAL FAT: LOCATION & FUNCTION

Liver
Stomach
Greater Omentum
Mesenteric Fat
Small Intestine
Peritoneal Cavity

WHY VISCERAL FAT MATTERS

Visceral Adipose Tissue → ↑ FREE FATTY ACID FLUX → Hepatic VLDL Overproduction

Macrophage Infiltration & Cytokines

↑ Triglycerides
↓ HDL
↑ Small Dense LDL

↑ Glucose
↑ Blood Pressure
↑ Pro-inflammatory & Pro-thrombotic State

WAIST CIRCUMFERENCE: HOW TO MEASURE

Measure at the top of the iliac crest (hip bone) at the end of normal expiration. Use a non-stretch tape, snug but not compressing the skin.

Iliac Crest

WAIST CIRCUMFERENCE THRESHOLDS

Population	Men	Women
United States	≥ 102 cm (40 in)	≥ 88 cm (35 in)
Europid (IDF)	≥ 94 cm (37 in)	≥ 80 cm (31.5 in)
South & East Asian (IDF)	≥ 90 cm (35 in)	≥ 80 cm (31.5 in)

Using US cut-points in other populations can underdiagnose.

KEY TAKEAWAY: Excess visceral adiposity is the root driver of metabolic syndrome. Measuring waist circumference identifies patients at risk—even before labs are abnormal.

WHAT TO CHANGE

- 1 Visceral Adipose Tissue (VAT)**
the inner edge of the SUBCUTANEOUS fat layer / outer surface of the abdominal musculature (sagittal panel, top left)
MOVE TO the yellow fat band INSIDE the peritoneum (~20px further in, deep to muscle + peritoneal line), or the omental fat around the bowel
- 2 Peritoneal Cavity (top-left sagittal panel) [CHECK]**
the subcutaneous-fat/muscle junction, within the abdominal wall
MOVE TO inside the peritoneal lining, i.e. the space containing bowel/omentum ~15-20px further in
- 3 (unlabelled leader stub, no text at either end) [CHECK]**
floating white leader with cyan tip inside the small-bowel loops at approx x=197-215, y=434 of the sagittal panel
MOVE TO nothing - it is an orphan leader with no label
- 4 Liver (Visceral Fat: Location & Function panel)**
the STOMACH (greater curvature / fundus, left upper quadrant = image right)
MOVE TO the dark red smooth mass in the RIGHT upper quadrant = image LEFT
- 5 Stomach (Visceral Fat panel)**
omental/preperitoneal fat at the left flank, ~15px lateral to the stomach wall
MOVE TO the striated pink J-shaped stomach itself
- 6 Mesenteric Fat (Visceral Fat panel) [CHECK]**
the same continuous yellow flank/omental fat band that the "Greater Omentum" leader 30px above also lands on
MOVE TO the fat within the mesentery, between the small-bowel loops
- 7 Small Intestine (Visceral Fat panel)**
yellow omental/flank fat at the right margin of the cavity, ~18px lateral to the nearest bowel loop
MOVE TO the coiled red small-bowel loops in the centre of the panel

NOTES

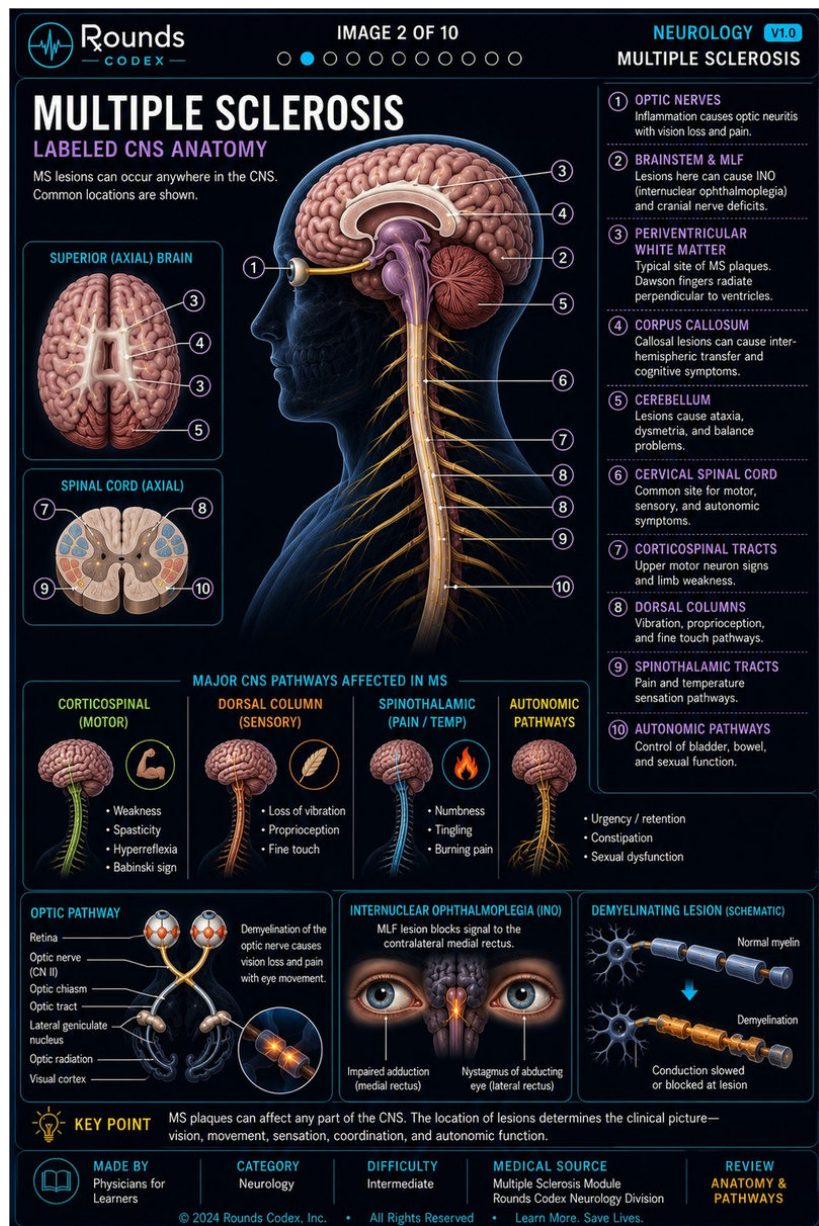
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ms-p2

Labeled CNS Anatomy · ms-02.jpg · 1024×1536 px · 8 findings

NOT MEASURED BY US — 8 of 30 labels flagged — 3 to move · 5 to check

The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.



WHAT TO CHANGE

- 2 - Brainstem & MLF (main sagittal figure)**
a gyrus of the OCCIPITAL/parietal cerebral cortex at the back of the hemisphere
MOVE TO the purple midline brainstem (midbrain/pons/medulla) in the centre of the figure
- 1 - Optic nerves (main sagittal figure) [CHECK]**
the eyeball (cornea/anterior globe)
MOVE TO the yellow optic nerve running back from the globe, about 19px to the right
- 3 - Periventricular white matter (main sagittal figure) [CHECK]**
the superior surface of the corpus callosum, the same white arc that label 4 marks 33px lower
MOVE TO the white matter around the ventricle, distinguishable from 4's target
- 9 - Spinothalamic tracts (main sagittal figure) [CHECK]**
the leader stops in the dark paraspinous/vertebral space about 19px lateral to the cord; a separate small white dot sits on the cord edge, so the line may simply be occluded
MOVE TO the cord itself, as leaders 6, 7, 8 and 10 do
- 8 - Dorsal columns (SPINAL CORD AXIAL inset)**
the dorsal-horn GREY MATTER at the dorsolateral position - the exact mirror image of where leader 7 (corticospinal) lands on the other side
MOVE TO the pale posterior funiculi flanking the midline septum, about 19px medial
- 7 - Corticospinal tracts (SPINAL CORD AXIAL inset) [CHECK]**
dorsal-horn grey matter about 7px medial to the blue dorsolateral white-matter block
MOVE TO inside the blue dorsolateral (lateral corticospinal) block
- Optic chiasm (OPTIC PATHWAY panel)**
the left grey-blue OPTIC TRACT, well below the crossing
MOVE TO the yellow X where the two optic nerves cross, about 27px up and to the right
- Lateral geniculate nucleus (OPTIC PATHWAY panel) [CHECK]**
forked leader - the upper branch correctly touches the left LGN, but the lower branch runs down-right and ends on the optic radiation fibre bundle
MOVE TO both branches on LGN tissue (left and right LGN)

NOTES

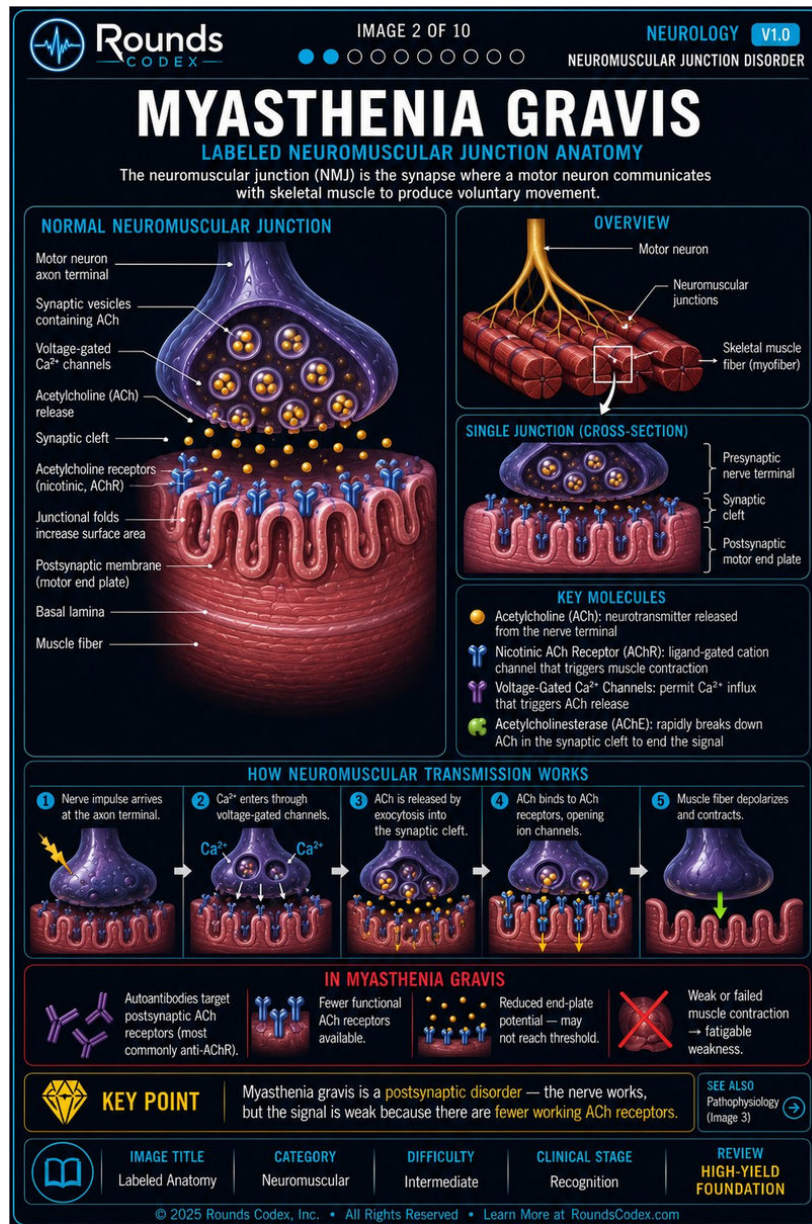
The order's FINDINGS have held up on every page we checked. Its DISTANCES have not: '~250px' measured 57 px on hyponatremia and '~160px' measured 35 px on cirrhosis. Work from the description and the source file, not from the order's numbers.

myasthenia-p2

Labeled Neuromuscular Junction Anatomy · myasthenia-02.jpg · 1024×1536 px · 4 findings

NOT MEASURED BY US — 4 of 18 labels flagged — 1 to move · 3 to check

The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.



WHAT TO CHANGE

- 1 Basal lamina (Normal Neuromuscular Junction panel)**
striated sarcoplasm deep inside the MUSCLE FIBER, about 58px below the base of the junctional folds and only 40px above where "Muscle fiber" points
MOVE TO the thin layer at the synaptic surface, within the cleft / lining the junctional folds
- 2 Voltage-gated Ca^{2+} channels [CHECK]**
presynaptic cytoplasm just inside the terminal's lateral membrane; no channel structure is drawn at the endpoint
MOVE TO in the presynaptic membrane (as the step-2 " Ca^{2+} " arrows below correctly do)
- 3 Acetylcholine (ACh) release [CHECK]**
plain dark background about 15px to the left of the terminal's lower edge, not on a fusing vesicle or a released ACh sphere
MOVE TO the vesicles fusing at the terminal's base, or the orange ACh in the cleft
- 4 Acetylcholine receptors (nicotinic, AChR) [CHECK]**
two leaders; the lower one correctly reaches a blue receptor, but the upper one passes OVER the receptor and terminates on an orange ACh molecule
MOVE TO both on the blue receptor glyphs (unless the upper is deliberately showing ACh bound)

NOTES

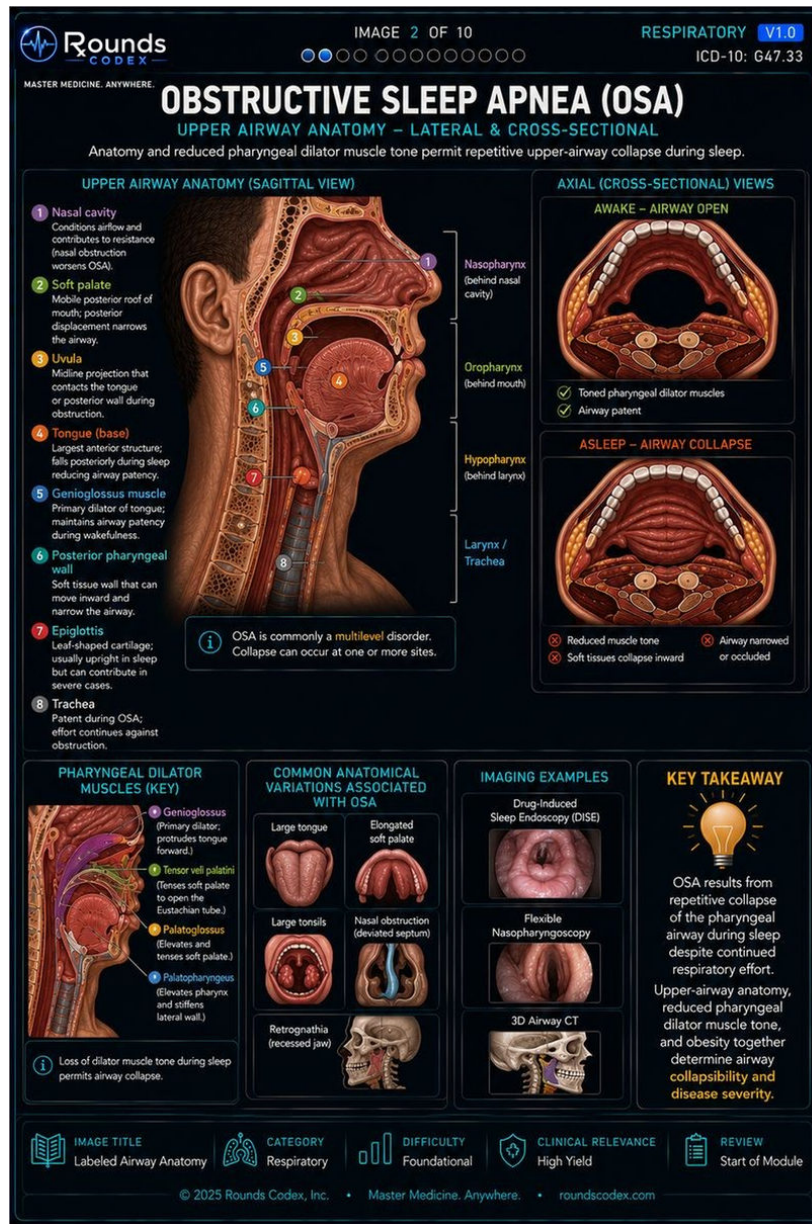
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osa-p2

Upper Airway Anatomy – Lateral & Cross-Sectional · osa-02.jpg · 800×1200 px · 6 findings

NOT MEASURED BY US — 6 of 16 labels flagged — 1 to move · 5 to check

The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.



WHAT TO CHANGE

- 2 Soft palate**
the nasal cavity floor above the mid hard palate (the leader runs to the right, away from the palate's mobile part)
MOVE TO the soft palate flap immediately below-left of the 2 marker
- 3 Uvula [CHECK]**
the dark oropharyngeal air space to the right of the uvula, ~30px off
MOVE TO the downward tip at the free posterior edge of the soft palate, which the 3 circle itself sits beside
- 5 Genioglossus muscle [CHECK]**
the posterior tongue border / palatoglossal fold region
MOVE TO the fan of muscle inside the tongue body running to the mandible
- 8 Trachea [CHECK]**
the pretracheal soft tissue of the anterior neck, having crossed over the tracheal rings
MOVE TO the ringed grey tube itself
- Palatoglossus (dilator key) [CHECK]**
the purple muscle band, which is the colour keyed to Genioglossus; no orange-keyed structure is drawn
MOVE TO an orange-highlighted palatoglossus
- Palatopharyngeus (dilator key) [CHECK]**
the same purple band
MOVE TO a blue-highlighted palatopharyngeus in the lateral pharyngeal wall

NOTES

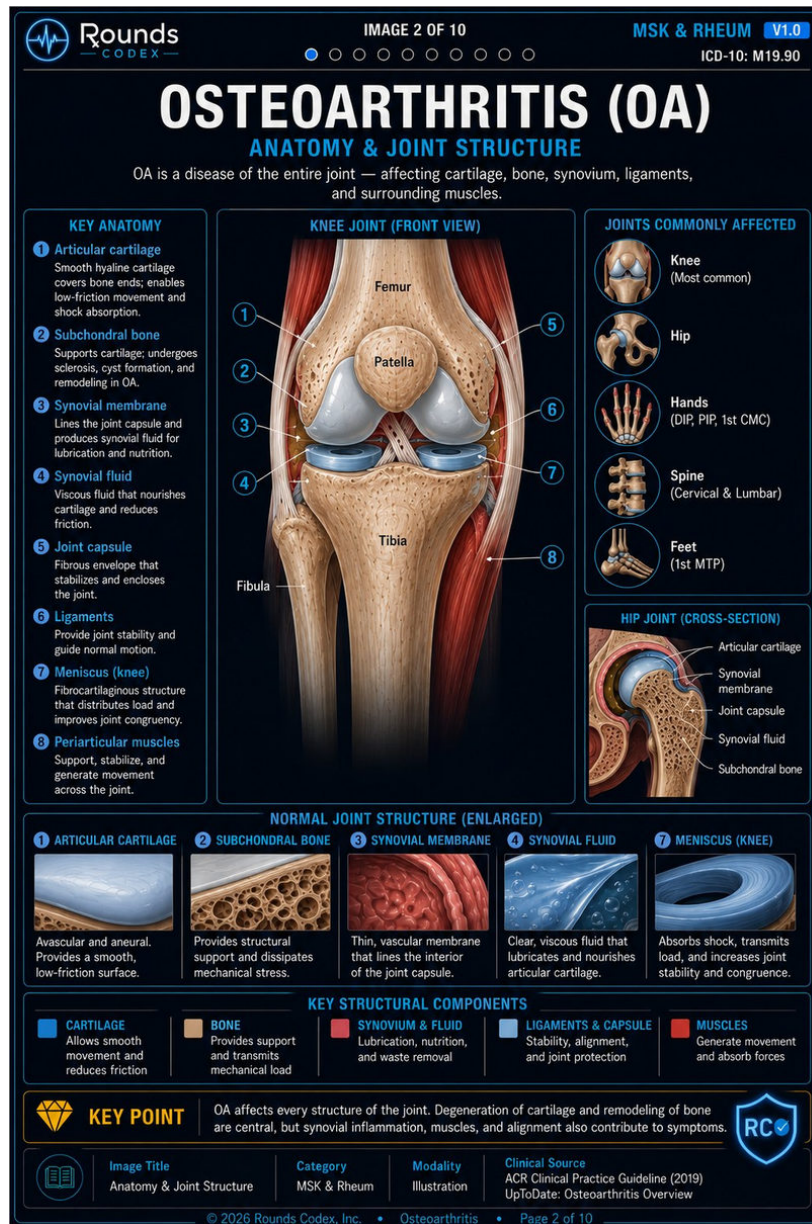
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osteoarthritis-p2

Anatomy & Joint Structure · osteoarthritis-02.jpg · 1024×1536 px · 7 findings

NOT MEASURED BY US — 7 of 17 labels flagged — 3 to move · 4 to check

The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.



WHAT TO CHANGE

1 1 Articular cartilage (KNEE JOINT FRONT VIEW)

an arrowhead squarely inside the tan cancellous/cortical bone of the distal femoral metaphysis, ~65 px above and medial to any cartilage

MOVE TO the grey-white articular cartilage capping the femoral condyle, lower and to the right

2 4 Synovial fluid (KNEE) [CHECK]

the top-left corner of the BLUE MENISCUS disc - the structure marker 7 names on the other side

MOVE TO the joint cavity/recess itself; arguable, since the tip sits in the crevice between condyle and meniscus where fluid would be

3 6 Ligaments (KNEE) [CHECK]

the gold/ochre peri-articular tissue lateral to the femoral condyle - the same band marker 3 "Synovial membrane" points to on the opposite side

MOVE TO a discrete ligament; the pale striated fibrous band ~60 px further lateral is what marker 5 "Joint capsule" uses

4 Synovial membrane (HIP JOINT CROSS-SECTION) [CHECK]

two branches, one ending in femoral-head cancellous bone at approx (827,844) after crossing straight over the salmon peri-articular layer, the other on the blue cartilage rim

MOVE TO the salmon/pink layer it crosses, which is the only membrane drawn

5 Joint capsule (HIP)

a bright dot inside the femoral-neck trabecular bone, ~12 px inside the cortex, with no capsule drawn anywhere near the endpoint

MOVE TO the salmon/pink band wrapping the joint superolaterally

6 Synovial fluid (HIP)

femoral-neck trabecular bone at approx (843,888), ~40 px below the joint space
MOVE TO the joint cavity between the acetabular and femoral cartilage

7 Subchondral bone (HIP) [CHECK]

mid-femoral-neck trabecular bone at approx (836,916)

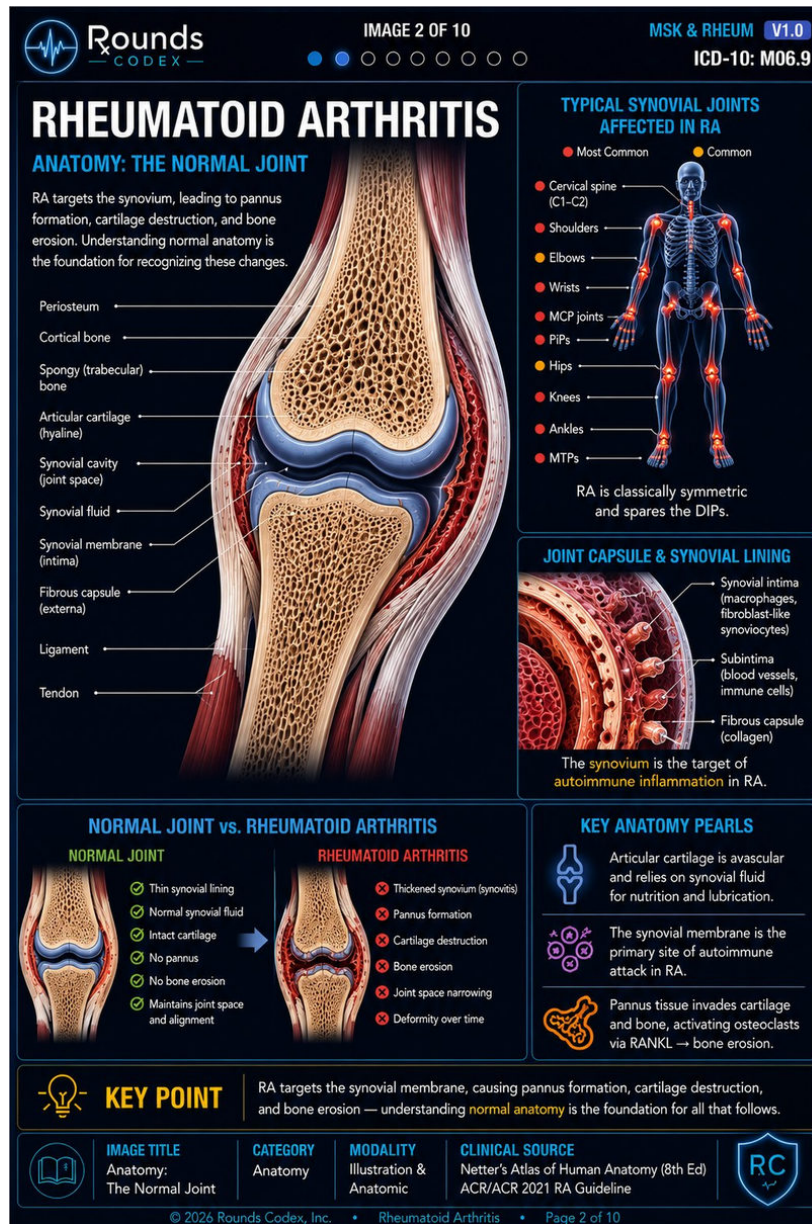
MOVE TO the dense plate immediately beneath the articular cartilage, ~60 px higher

NOTES

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NOT MEASURED BY US — 7 of 23 labels flagged — 3 to move · 4 to check

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WHAT TO CHANGE

- Hips (Typical synovial joints panel)**
the LEFT KNEE - the leader runs horizontally at knee level and touches the femoral condyle of the glowing knee joint
MOVE TO the glowing hip joints ~80 px higher, at the femoral heads
- Knees (Typical synovial joints panel)**
the proximal tibia / calf soft tissue about 30 px BELOW the glowing knee joint, in plain blue leg
MOVE TO the knee joint glow itself (which the "Hips" leader is currently touching)
- Tendon (main joint figure)**
a dot squarely inside the striated RED MUSCLE BELLY
MOVE TO the white fibrous tendon band ~20 px up and to the right of the dot (the "Ligament" leader lands on it correctly)
- Synovial fluid [CHECK]**
the red synovial lining tissue at the joint margin, ~34 px left of where the "Synovial cavity" arrow lands
MOVE TO inside the joint space / recess that holds the fluid
- Synovial membrane (intima) [CHECK]**
the margin of the tibial articular cartilage, after the leader has crossed straight over the red synovial lining it names
MOVE TO the red synovial lining itself (arguable - the synovium does reflect at that cartilage margin)
- Fibrous capsule (externa) (main figure) [CHECK]**
the tibial cortical bone at the cartilage-bone junction, after crossing over the pale fibrous capsule layer
MOVE TO the pale outer fibrous layer at approx x=250-272 (arguable - the capsule inserts on bone there)
- Fibrous capsule (collagen) (Joint capsule & synovial lining panel) [CHECK]**
deep red synovial tissue between two villi, having crossed straight over the pale layered collagen band at the outer edge
MOVE TO that pale outer collagen band

NOTES

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sickle-cell-p2

Anatomy: Hemoglobin, RBCs, Spleen & Microvasculature · sickle-cell-02.jpg · 1024×1536 px · 5 findings

NOT MEASURED BY US — 5 of 11 labels flagged — 1 to move · 4 to check

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IMAGE 2 OF 10

HEME & ONC V1.0

ICD-10 D57.1

SICKLE CELL DISEASE

ANATOMY: HEMOGLOBIN, RBCs, SPLEEN & MICROVASCULATURE

Know the key structures and how they work—so the disease makes sense.

HEMOGLOBIN (Hb)

Oxygen-carrying protein made of 4 polypeptide chains.

α chain
β chain

SICKLE MUTATION (HbS)

β-globin: Glutamic acid (Glu) → Valine (Val) at position 6

Glu (E) Val (V)

A single amino-acid change causes HbS to polymerize when deoxygenated.

RED BLOOD CELLS (RBCs)

NORMAL RBC

Biconcave disc, flexible and deformable.

SICKLED RBC

Rigid, crescent-shaped, less deformable.

Normal RBCs flow freely

Sickled RBCs cause vaso-occlusion & ischemia

SPLEEN: FILTER & PROTECT

Filters abnormal cells and removes old RBCs. In sickle cell disease, spleen dysfunction and infarction are common.

Capsule
Trabeculae
White pulp (immune)
Red pulp (filters blood)
Splenic artery
Splenic vein

HEMOLYSIS (CHRONIC)

Sickled and damaged RBCs are destroyed, leading to anemia and jaundice.

↑ Indirect bilirubin
↑ LDH
↓ Haptoglobin
↑ Reticulocytosis

MICROVASCULATURE & ISCHEMIA

Sickled RBCs adhere to endothelium, causing blockage, reduced flow, and tissue ischemia.

Endothelium Adhesion molecules Sickled RBCs

Normal Flow Obstructed Flow

Oxygen (Normal) Low Oxygen (Hypoxia)

Ischemia & Tissue Injury

MAJOR ORGANS AFFECTED

- Brain — Stroke risk
- Lungs — Acute chest syndrome
- Heart — Pulmonary hypertension
- Spleen — Infarction, sequestration
- Kidneys — Papillary necrosis
- Bone — Painful crises (osteonecrosis)
- Liver/GB — Gallstones, jaundice
- Penis — Priapism
- Eyes — Retinopathy (less common)

THE SICKLE PROCESS: A SMALL CHANGE, A BIG IMPACT

Soluble hemoglobin → HbS polymerizes → Rigid, crescent shape → Blocks blood flow → Pain & tissue injury → RBC destruction → Anemia & jaundice

KEY TAKEAWAY

A single amino acid change in β-globin creates HbS. When deoxygenated, it polymerizes, causing sickling, vaso-occlusion, and hemolysis—leading to lifelong complications.

10

IMAGE TITLE

Labeled Anatomy

CATEGORY

Hematology & Oncology

MODALITY

Illustration

CLINICAL SOURCE

Hematology Guidelines • UpToDate • ASH • NHLBI

REVIEW

CLINICAL PEARLS

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WHAT TO CHANGE

- Trabeculae**
the right margin of the uppermost lavender white-pulp follicle, a few px into red pulp; there is no trabecular band anywhere near the endpoint
MOVE TO one of the pale fibrous bands running inward from the capsule
- Red pulp (filters blood) [CHECK]**
the right margin of the lowest lavender white-pulp follicle, 1-2 px outside it - visually identical to how the "White pulp" leader one row above terminates on the middle follicle
MOVE TO the dark red inter-follicular parenchyma, which fills most of the organ and is untouched
- Splenic vein [CHECK]**
the pink capsule at the inferomedial corner of the spleen, after a dogleg that turns UP away from the vessels
MOVE TO the blue splenic vein drawn about 20 px to its left at the hilum, the way "Splenic artery" reaches the red vessel
- Endothelium [CHECK]**
a dark red blood cell inside the lumen, about 19 px past the pale endothelial lining the line crosses on its way down
MOVE TO the pale lining band itself
- Adhesion molecules [CHECK]**
a bright red blood cell in the lumen, about 12 px below the vessel wall where the small dark adhesion molecules are drawn (the line passes one on its way past)
MOVE TO one of those dark wall-bound molecules

NOTES

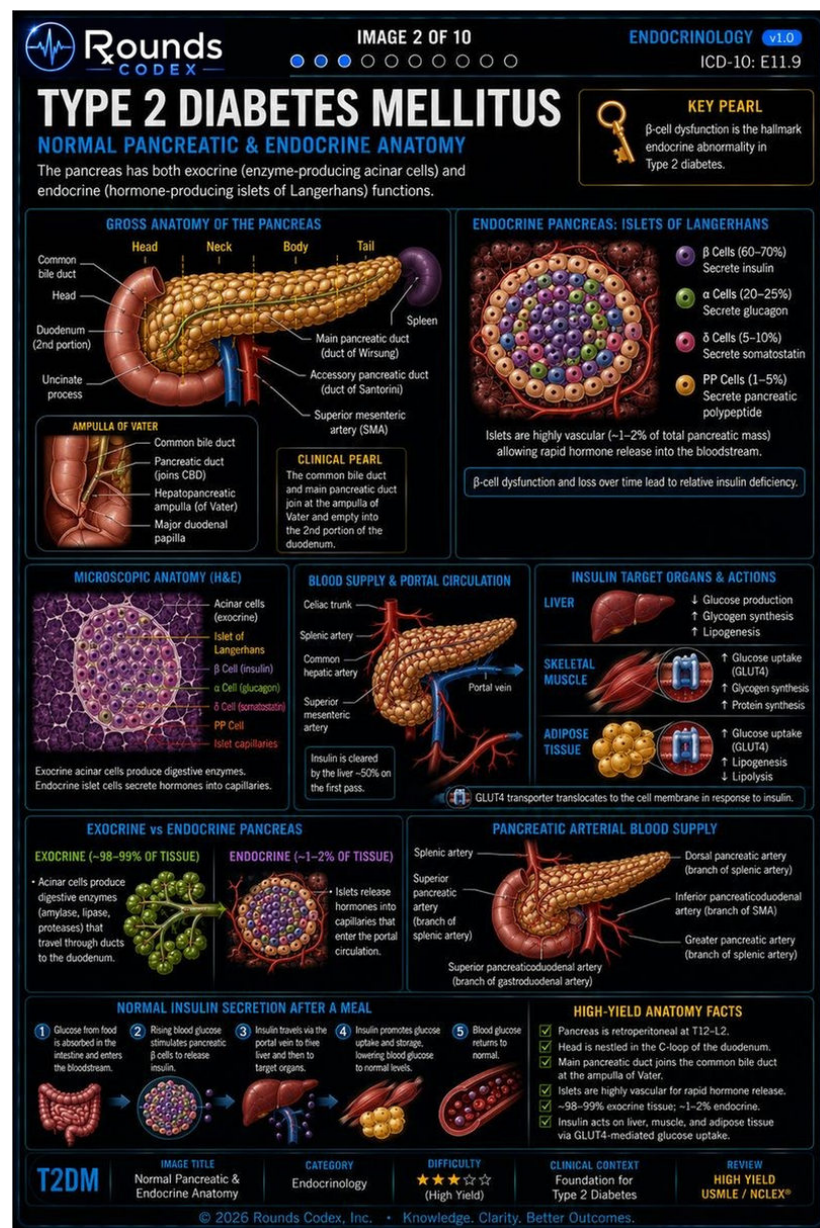
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t2dm-p2

Normal Pancreatic & Endocrine Anatomy · t2dm-02.jpg · 800×1200 px · 6 findings

NOT MEASURED BY US — 6 of 30 labels flagged — 2 to move · 4 to check

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WHAT TO CHANGE

- Accessory pancreatic duct (duct of Santorini) [Gross Anatomy panel]**
the horizontal arterial branch of the red SMA trunk at (244,338), below and outside the pancreas - no duct at the endpoint
MOVE TO the small duct in the upper pancreatic head, superior to the duct of Wirsung
- Greater pancreatic artery (branch of splenic artery) [Pancreatic Arterial Blood Supply panel]**
the tip of an inferior branch of the SMA at (605,908), below the pancreatic head - the same arcade the Inferior pancreaticoduodenal label points into
MOVE TO a branch off the splenic artery entering the pancreatic body along its superior border
- Dorsal pancreatic artery (branch of splenic artery) [Arterial Blood Supply panel] [CHECK]**
the parenchyma at the very tip of the pancreatic tail (635,830); no artery resolvable at the endpoint
MOVE TO near the pancreatic NECK, where the dorsal pancreatic artery leaves the splenic
- Superior pancreatic artery (branch of splenic artery) [Arterial Blood Supply panel] [CHECK]**
the outer wall of the duodenum at (483,869), at most on the thin pancreaticoduodenal arcade vessel running over it
MOVE TO the superior border of the pancreatic body
- Common bile duct [Gross Anatomy panel] [CHECK]**
an arrowhead on the medial wall of the descending duodenum at (117,277)
MOVE TO the pale branching duct structure ~10-12 px to the right at (128,283), where the duct enters the pancreatic head. Note: the adjacent "Head" leader DOES continue past the duodenum to the gland, so the short stop here is not a drawing convention
- Splenic artery [Blood Supply & Portal Circulation panel] [CHECK]**
a branch descending down-LEFT from the celiac trunk at (352,614), the same descending trunk the "Common hepatic artery" leader reaches ~23 px lower
MOVE TO the vessel that leaves the celiac to the upper right and runs along the superior border of the pancreas to the tail

NOTES

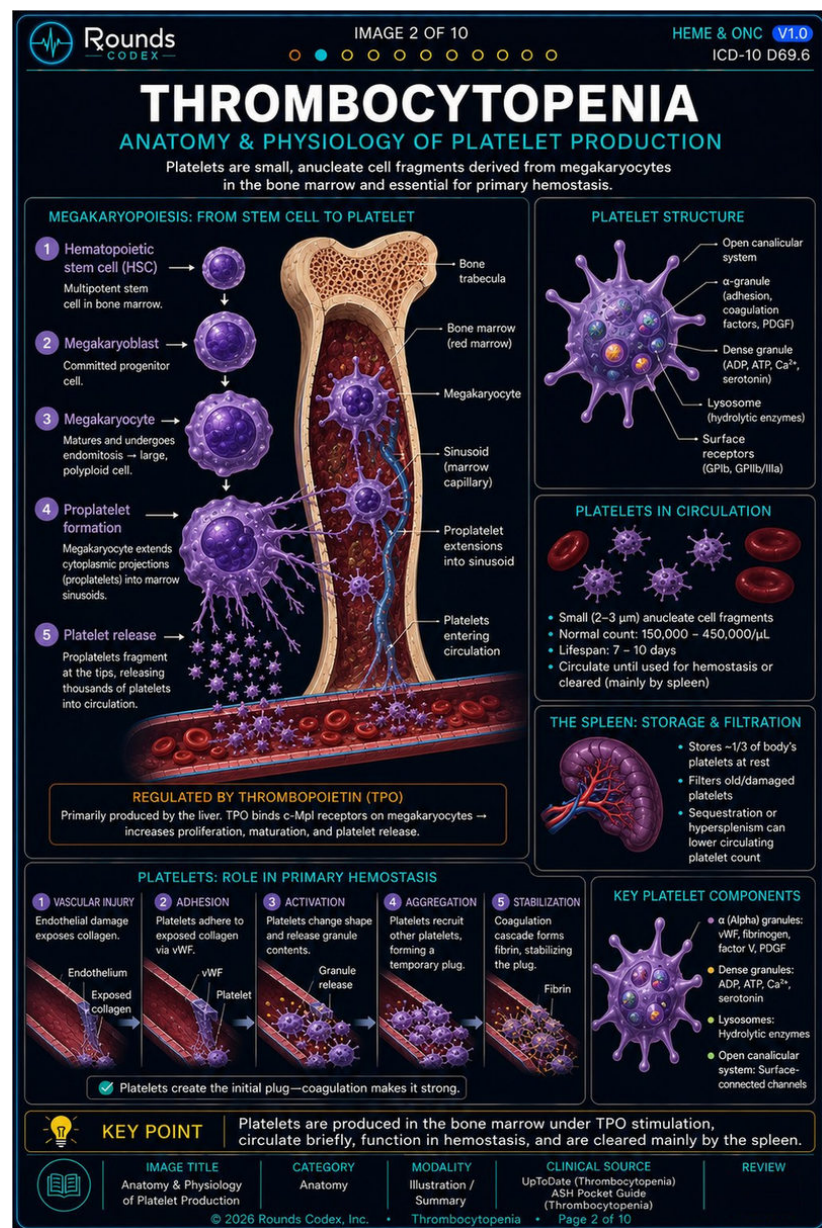
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thrombocytopenia-p2

Anatomy & Physiology of Platelet Production · thrombocytopenia-02.jpg · 1024×1536 px · 5 findings

NOT MEASURED BY US — 5 of 22 labels flagged — 1 to move · 4 to check

The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.



WHAT TO CHANGE

- Sinusoid (marrow capillary)**
a purple megakaryocyte pseudopod (a proplatelet extension); the leader doglegs, crosses the blue sinusoid it names, and carries on about 15 px past it before stopping on the purple process
MOVE TO the blue sinusoid where it crosses it
- Proplatelet extensions into sinusoid [CHECK]**
plain red marrow about 6 px outside the right wall of the blue sinusoid, with no purple process at the tip; the nearest proplatelet is ~25 px to the left
MOVE TO one of the purple cytoplasmic strands entering the vessel
- Lysosome (hydrolytic enzymes) [CHECK]**
the lower border of the small blue-green granule at the 3-o'clock position - the same granule the "Dense granule" leader is aimed at from the other side
MOVE TO a granule of its own; the two leaders cannot both be right
- Dense granule (ADP, ATP, Ca²⁺, serotonin) [CHECK]**
plain purple cytoplasm at the base of a pseudopod, about 7 px right of that same blue-green granule
MOVE TO a granule; note the two orange-colored granules in the platelet carry no label at all
- Open canalicular system [CHECK]**
dark background about 5 px beyond the tip of the upper-right pseudopod bulb, outside the cell
MOVE TO the internal channel network drawn inside the platelet body

NOTES

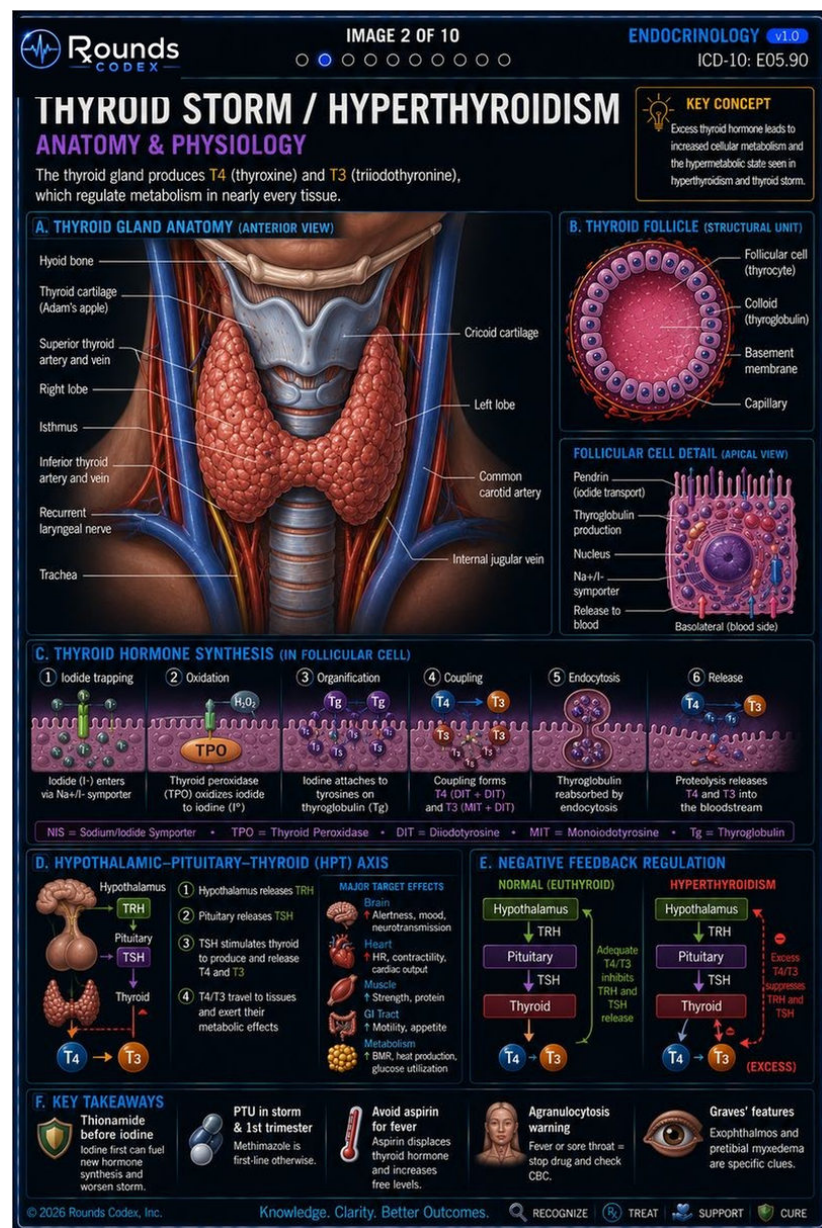
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thyroidstorm-p2

Anatomy & Physiology · thyroidstorm-02.jpg · 800×1200 px · 10 findings

NOT MEASURED BY US — 10 of 21 labels flagged — 5 to move · 5 to check

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WHAT TO CHANGE

- Cricoid cartilage**
the grey THYROID cartilage lamina, at almost the same height as the "Thyroid cartilage (Adam's apple)" leader; the cricoid ring is the separate narrower band about 45 px lower
MOVE TO that lower ring
- Trachea**
the gold nerve band running down the lateral neck; a red artery and ~31 px of soft tissue separate the tip from the grey ringed trachea
MOVE TO the ringed tube in the midline
- Common carotid artery**
the middle of the blue internal jugular vein; the leader crosses the red artery on its way in and stops ~11 px past it
MOVE TO that red artery
- Follicular cell (thyrocyte) (panel B)**
the pink colloid, about 16 px past the inner border of the epithelial ring it crosses - i.e. in the compartment the "Colloid (thyroglobulin)" label names
MOVE TO one of the cuboidal cells in the ring
- Nucleus (Follicular Cell Detail)**
a small dark purple cytoplasmic vesicle at the cell's left margin, about 21 px short of the large purple nucleus
MOVE TO the nucleus
- Superior thyroid artery and vein [CHECK]**
the gold nerve band at the medial edge of the blue jugular; no thyroid artery or vein at the tip
MOVE TO the vessels running to the superior pole
- Inferior thyroid artery and vein [CHECK]**
the lateral border of the thyroid lobe immediately beside the gold nerve band, with no vessel at the tip
MOVE TO the red/blue vessels at the lower pole
- Internal jugular vein [CHECK]**
the gold vagus nerve about 6 px medial to the blue vein
MOVE TO the blue vein
- Capillary (panel B) [CHECK]**
the basal side of the follicular epithelium, about 12 px inside the orange basement membrane the leader crosses; the capillary plexus it names is outside that ring
MOVE TO one of the dark red vessels outside the ring

NOTES

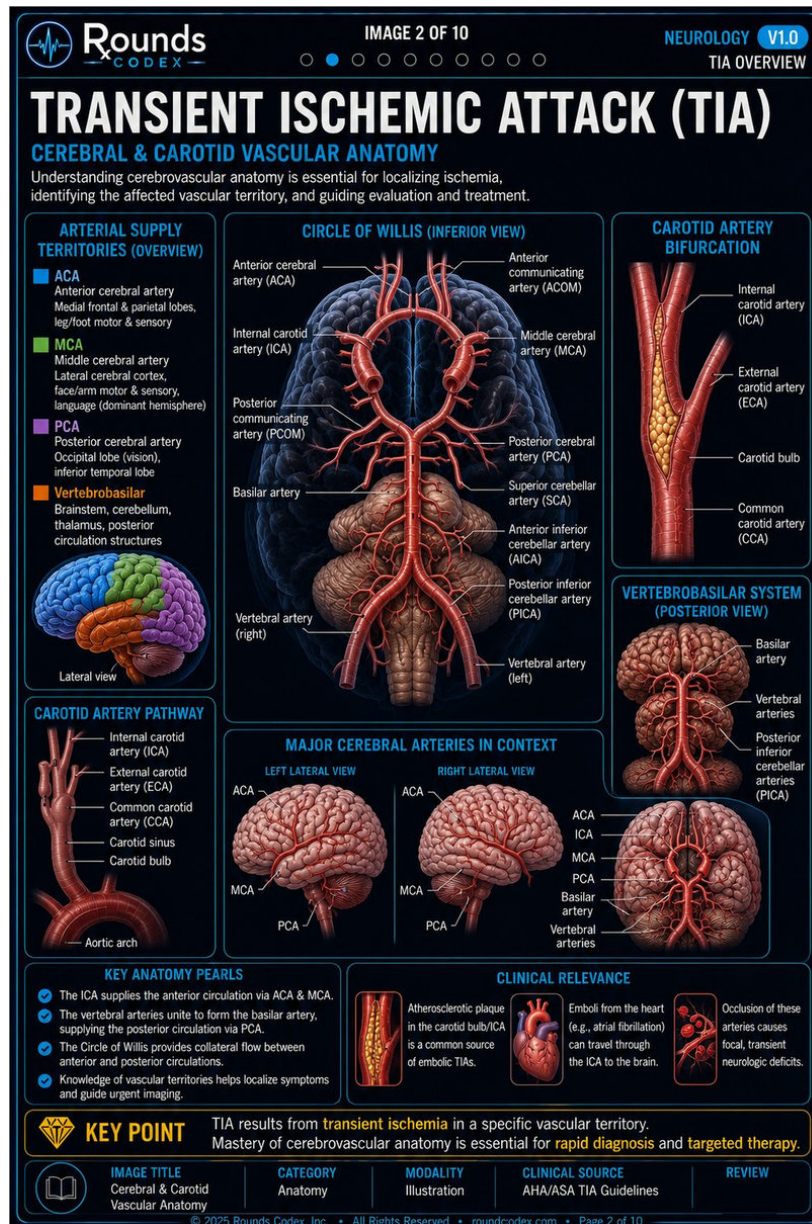
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tia-p2

Cerebral & Carotid Vascular Anatomy · tia-02.jpg · 1024×1536 px · 9 findings

NOT MEASURED BY US — 9 of 37 labels flagged — 5 to move · 4 to check

The rows below are the WORK ORDER'S text, reproduced as written. No endpoint on this page has been measured and no tissue sampled. Treat it as a candidate list, not as evidence.



WHAT TO CHANGE

- Internal carotid artery (ICA) (Circle of Willis)**
the rounded lateral end of the left M1 segment - the mirror image of exactly what the "MCA" label points at on the other side
MOVE TO the thick trunk with the open cut lumen below the bifurcation, ~40px to the right
- Anterior communicating artery (ACOM) [CHECK]**
the proximal ascending (A2) segment of the right ACA, ~21px above the midline connector
MOVE TO the short horizontal midline vessel joining the two A1 segments
- Basilar artery (Circle of Willis) [CHECK]**
cerebellar surface ~9px short of the trunk's left wall, and a second arm of the same leader runs up to a lateral cerebellar/PCA branch
MOVE TO the midline vertical trunk
- Common carotid artery (CCA) (carotid pathway panel)**
the dilated carotid bulb
MOVE TO the straight tube below the bulb
- Carotid bulb (carotid pathway panel)**
the straight common carotid ~45px below the dilated segment
MOVE TO the ovoid swelling, which the CCA label currently occupies
- Carotid sinus (carotid pathway panel)**
the straight common carotid ~23px below the dilated segment
MOVE TO the dilated segment - all three labels in this stack are shifted one step down
- Vertebral arteries (vertebrobasilar posterior view)**
a small vessel on the cerebellar surface mid-panel
MOVE TO the paired ascending trunks at the bottom of the panel that fuse into the basilar, ~48px lower
- ACA (vertebrobasilar lower sub-panel) [CHECK]**
left frontal cortex, fading out ~27px short of the vessel
MOVE TO the anterior midline vessels of the circle
- PCA (left lateral view, arteries in context) [CHECK]**
the anterior brainstem where the basilar/vertebral lies
MOVE TO the vessel curving around the midbrain toward the occipital lobe; the right lateral view looks the same way

NOTES

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